Reliability Engineering and System Safety 278 (2027) 113521



Contents lists available at ScienceDirect

Reliability Engineering and System Safety

journal homepage: www.elsevier.com/locate/ress



Cost-of-unreliability decomposition and airworthiness review register from linked part 145 maintenance records

Antonio C.P. Mesquita

Aerospace Logistics Engineering Laboratory (AeroLogLab), Instituto Tecnológico de Aeronáutica (ITA), São José dos Campos, 12228-900, SP, Brazil

ARTICLE INFO

Keywords:
Cost of unreliability
Continuing airworthiness
Age replacement
Censored data
Recurrent events
Spare parts provisioning
Maintenance records

ABSTRACT

Unscheduled component removals generate material cost and service disruption in certificated fleets, but count-based reviews can overlook infrequent, economically consequential events. Existing workflows also separate subsystem ranking, replacement timing, and spares decisions, leaving economic evidence and approved safety action unlinked. We developed a cost-of-unreliability (COUR) workflow linking Part 145 maintenance records to a non-compensatory continuing-airworthiness review register with two arms: a safety arm recording an operator-approved disposition, and an economic arm combining chapter-level COUR with censoring-aware survival and age-replacement analysis, recurrent-event burden, demand dispersion, and repairable-stock benefit–cost screening. We evaluated it on a complete offshore medium twin-engine helicopter fleet, reconstructing right-censored episodes and delayed entry from linked replacement histories and propagating each component's AIC-selected distribution into its cost-rate calculation. Under the mid grounding-cost scenario, fleet COUR totalled $92.21 million. The strongest age-replacement candidate yielded a modelled 68.2% saving; components whose selected model had no finite cost-rate minimum were excluded from candidacy. Hold-out COUR ranks agreed strongly (ρ = 0.903) and retained the leading chapters, although mid-tier priorities moved. Because grounding duration, safety dispositions, and post-removal findings were unavailable, outputs remain review candidates, not approved actions. The register lets continuing-airworthiness and supply teams coordinate economic interventions auditably, without cost overriding safety.

1. Introduction

Unscheduled component removals impose parts and corrective-labour costs on certificated fleets and may expose operators to aircraft-on-ground (AOG) losses. Reliability reviews nevertheless often rank subsystems by removal volume alone. This practice can divert engineering attention from low-frequency removal streams with high economic burden, a familiar gap in operational reliability assessment for transportation-critical systems [1,2]. Certificated helicopter operators face the problem acutely because AOG events interrupt revenue service, yet the administrative records needed for economic re-ranking are already produced under European Union Aviation Safety Agency (EASA) continuing-airworthiness rules and maintenance-organisation obligations [3]. This study develops and evaluates a bounded decision-support workflow on 24 offshore medium twin-engine helicopters (OMTH) over 27 months.

Problem. In practice, reliability reviews still begin with removal *volume*: Air Transport Association (ATA) chapters with the highest event counts receive engineering and stocking attention first. Removal counts measure maintenance activity, not demonstrated aircraft-on-ground risk, yet volume ranking misrepresents economic burden: a chapter ranked twenty-fifth by count can rank eighth under cost of unreliability (COUR), crossing 17 higher-volume chapters, so economically consequential low-frequency systems remain hidden. The consequence is misallocation across three supportability decisions (engineering-focus priorities, scheduled-replacement candidates, and incremental stock), each resolved on separate, economically inconsistent criteria from the same administrative records.

Relevance. If the hidden-system problem is not resolved, engineering attention and stocking resources continue to flow to high-count, low-cost chapters while systems with disproportionate economic burden accumulate unchecked corrective cost and AOG exposure. Scheduled-replacement candidates and high benefit–cost ratio (BCR) repairables remain unidentified, so operators pay repeatedly for unscheduled removals that structured economic triage would flag for intervention. The evidence needed to surface these priorities is already captured under Part 145 obligations; the gap is analytical, not data. For continuing airworthiness management, volume-only triage can likewise misorder review priority: count-based screens do not aggregate the parts and

E-mail address: celio@ita.br.

https://doi.org/10.1016/j.ress.2026.113521
Received 20 June 2026; Received in revised form 14 August 2026; Accepted 11 September 2026
Available online 21 September 2026
0951-8320/© 2026 The Author. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).
The Article Processing Charge for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614).

labour signals that COUR uses to rank economically consequential systems. COUR nevertheless measures economic burden, not the probability or severity of a safety consequence; the economic screen must therefore remain non-compensatory and subordinate to the operator's approved safety disposition.

Approach and research questions. The economic arm produces three outputs: engineering-focus priorities, scheduled-replacement candidates, and incremental-stock candidates. An external, operator-approved safety disposition forms the separate safety arm and always takes precedence. COUR first decomposes fleet corrective burden into parts, work-order-linked labour, and scenario-bounded AOG opportunity cost, then re-ranks ATA priorities. After COUR re-ranking, components that pass the wear-out screen ($\beta_w > 1.05$) are compared under Weibull, log-normal, and exponential models; the AIC-selected distribution then enters cost-rate optimisation [2]. The survival stage is complemented by a negative-binomial (NB) panel estimate of fleet-level demand dispersion (α) for METRIC stress testing and, separately, exploratory usage-intensity effects after fixed effects. The stock stage applies a Poisson-demand METRIC screen that ranks repairables by benefit–cost ratio (BCR) [4]. Outputs are bounded prescriptive candidates for engineering review, not approved programme changes. The empirical base comprises 33,486.6 fleet flight hours and 1711 tracked unscheduled mechanical removals (January 2024–March 2026). Five research questions guide the study:

1. **Economic priority.** How materially does COUR change ATA rankings, and which systems (potentially under-screened by volume triage) diverge most in parts–labour–grounding economic burden?
2. **Fleet burden.** What are total COUR, its parts–labour–AOG decomposition, and its sensitivity band?
3. **Candidate actions.** Which wear-out components merit T^* review, and which incremental stocks rank highest by BCR?
4. **Normalised burden.** What are COUR and removal intensity per 1,000 flight hours and aircraft-year?
5. **Priority stability.** Do exposure-adjusted removal and COUR priorities persist over a six-month hold-out?

The paper makes three contributions to operational reliability decision support. First, a COUR decomposition from linked maintenance records separates parts, labour, and scenario-bounded grounding exposure, with bootstrap assessment and exposure-normalised indicators. It re-ranks fleet subsystem priorities by up to 17 positions relative to removal volume and provides an exact pairwise criterion for when per-removal cost heterogeneity overturns count separation. Second, an integrated pipeline couples Weibull cost-rate screening, negative-binomial demand calibration, and METRIC benefit–cost triage under explicit scenario bounds, with prospective hold-out validation of both removal priorities ($\rho = 0.887$) and COUR ranks ($\rho = 0.903$). Third, the outputs populate the economic arm of a non-compensatory continuing-airworthiness review register, in which an operator-approved safety disposition is a separate, precedence-bearing input rather than a quantity inferred from COUR.

Section 2 reviews related work and states the research gap. The methods that follow in Section 3 cover the data, the COUR decomposition, survival and METRIC screening, and the hold-out design. Results appear in Section 4, ordered as removal burden, COUR re-ranking and sensitivity, prescriptive candidates, hold-out stability, and the review register itself. Section 5 interprets that evidence and its limits; Section 6 closes with findings and implications.

2. Background

Prior work is organised around five dimensions that structure the workflow: volume Pareto ranking, cost-of-unreliability (COUR) re-ranking, linked labour burden, survival modelling of time to removal, and bounded prescriptive outputs (age-replacement intervals T^* and incremental stock benefit–cost ratios). Prospective hold-out evaluation runs across these dimensions rather than distinguishing them, and is treated as a reporting requirement instead of a point of comparison. Operational reliability programmes increasingly use field reliability databases [5,6], yet transportation fleets rarely connect survival evidence, event-cost decomposition, and bounded maintenance-policy outputs in one auditable workflow [7].

Volume ranking and survival-based replacement. ATA-level Pareto analysis remains the default starting point for fleet reliability reviews, but removal count alone understates systems with fewer yet costlier events. Rotorcraft studies have monitored ATA-chapter trends [8], estimated component survival from in-service records [9,10], and optimised component planning [11], but they do not link survival estimates to fleet-level COUR ranking, labour burden, and repairable stocking. Within *Reliability Engineering & System Safety*, replacement-policy and civil-aircraft risk studies [12,13] likewise do not connect economic ranking to labour burden or repairable stocking on matched operational records. Classical replacement theory [1,2,14] supplies the cost-rate framework used later for candidate intervals T^*; no cited rotorcraft study prospectively tests whether exposure-normalised economic priorities remain stable.

Labour burden from linked maintenance records. Survival evidence can identify age-dependent time-to-removal behaviour, but it does not establish the full corrective burden unless removal records are connected to execution effort. Applied maintenance, repair, and overhaul (MRO) analytics [6,15] highlights data accessibility and quality as barriers to fleet-level deployment; the rare-event stream motivates Pareto-style concentration analysis. The operator-level challenge is to link flight hours, removals, and work-order labour within one economic assessment, addressed here through ATA-level labour burden indicators from matched Part 145 records.

Repairable inventory and bounded stocking. Once corrective burden is measured, economic priority must be translated into bounded supply action. METRIC-type queueing approximations [4] underpin repairable spare-parts optimisation, while aviation and maintenance–inventory studies address irregular demand, fleet-level cost reduction, condition-based maintenance, and multi-item stock planning [7,16–21]. These studies define the theoretical state of the art but do not calibrate METRIC benefit–cost screening from linked Part 145 rotorcraft removal and work-order records, nor restrict outputs to incremental stock candidates under explicit COUR scenario bounds.

Removal economics and operational context. Removal economics provides the common basis for engineering focus, replacement timing, and stock. Helicopter lifecycle-cost work shows that reliability, maintainability, parts replacement, and maintenance overhaul dominate ownership and direct operating cost [22,23], but programme-level lifecycle cost (LCC) modelling does not yield the event-level parts–labour–AOG decomposition available from routine Part 145 records. Empirical work on event-level maintenance cost structure from linked operational records remains sparse; data-driven and degradation-optimisation studies [7,24,25] advance reliability modelling but do not close the loop from COUR through prescriptive age-replacement and incremental stock screening on linked rotorcraft administrative data.

Research gap and positioning. Three shortfalls motivate the analysis. First, no prior study integrates COUR ranking, labour burden, demand modelling, survival-based age-replacement candidates, METRIC benefit–cost screening, and prospective hold-out validation for a complete certificated fleet. Second, the bridge from survival-based cost-rate minima to scheduled-replacement templates and METRIC stock rankings has not been demonstrated under explicit grounding-scenario bounds. Third, volume-only triage misranks corrective burden. A volume top-10 screen omits ATA 72 (Engine; volume rank 25) and ATA 23

Table 1
Scope of selected cited works against removal-economics dimensions.

Study	Domain	*Pareto ranking*	*COUR decomp.*	*Labour burden*	*Survival modelling*	*Replacement policy*	*Spare stocking*	*Ordering diagnostic*	*Non-comp. register*
Apostolidis et al. [15]	Aviation	•		•					
Badía et al. [12]	Generic				•	•			
Ş.ahin et al. [16]	Aviation		•[a]				•		
Wang et al. [7]	Production		•[a]		•[b]	•	•		
Ghobbar and Mukherjee [17]	Aviation		•[a]				•		
Zheng et al. [20]	Production					•	•		
Guo et al. [13]	Aviation					•			
Duan et al. [25]	Aviation					•			
Present study	**Aviation**	•	•	•	•[c]	•	•	•	•

Table note. • = substantive treatment; blank = not addressed. *Domain* records the setting in which each method is demonstrated: *Aviation* for aircraft or fleet applications, *Production* for manufacturing or degrading production systems, and *Generic* for method or policy papers demonstrated without a sector-specific case. The present study is an aviation application on a certificated rotorcraft fleet; the two production-system comparators and the generic comparator therefore transfer as method evidence rather than as directly comparable operating contexts.

[a] Denotes inventory or lifecycle-cost emphasis rather than event-level parts–labour–AOG COUR from linked Part 145 records. *Ordering diagnostic*: exact pairwise condition under which per-removal cost heterogeneity overturns volume separation. *Non-comp. register*: non-compensatory fusion of an operator-approved safety disposition with economic evidence. Replacement policy and spare stocking are shown separately because most cited works address only one: age-replacement and condition-threshold studies set replacement or inspection timing, whereas inventory studies set stock levels. Policy classes also differ within the replacement column—the present study minimises an age-based cost rate, Badía et al. [12] compares age replacement with replacement after N minimal repairs, and the condition-based studies set degradation thresholds or inspection intervals.

[b] Wang et al. [7] models a degradation process rather than a time-to-removal distribution.

[c] The present study does not assume a single family: Weibull, log-normal, and exponential are fitted to right-censored, delayed-entry episodes and selected by AIC per component. Neither the observation that cost ranking differs from count ranking nor the use of non-compensatory rules is new; what is proposed here is the exact algebraic inversion condition and its embedding in a two-axis continuing-airworthiness register. Prospective hold-out evaluation is not tabulated because it is established practice rather than a discriminating dimension; it is reported in Sections 4.11 and 4.12. Programme-level helicopter LCC studies [22,23] and the offshore service-quality synthesis [27] are discussed in the review but omitted because they address ownership economics or service quality rather than record-based removal-economics closure. Applied MRO analytics cases [6] are discussed with Apostolidis et al. [15] but not shown as a separate row.

(Communications; volume rank 11) and yields no scheduled-replacement or incremental-stock register rows. The contribution integrates established estimators within a COUR-centred register and adds an exact ordering diagnostic derived from the COUR decomposition; it does not propose a new reliability estimator.

Two qualifications keep that comparison honest. First, prospective out-of-sample evaluation is long-standing practice in this literature and is therefore not tabulated as a discriminating dimension; the hold-out reported in Section 4 is distinctive in what it evaluates—an economic priority ranking over ATA chapters—rather than in the act of holding data out. Second, non-compensatory decision rules are established in maintenance and risk analysis: Certa et al. [26] classify failure modes by ELECTRE TRI outranking precisely because severity should not be compensated by other risk parameters. The present register adopts that principle and applies it to a different separation, namely an externally approved safety disposition versus record-based removal economics.

Table 1 summarises these empirical and joint-optimisation comparators chronologically.

Positioning of the ordering diagnostic. Within the selected literature in Table 1, field-data and MRO studies address reliability databases and execution evidence [6,15], while maintenance–inventory and degradation studies optimise replacement or stock decisions [7,17]. They do not provide a criterion for deciding when an event-count ranking is already economically order-equivalent to a linked-cost ranking. The diagnostic below fills that narrower gap: it is derived directly from the COUR identity, requires no additional estimator, and shows whether observed per-removal cost heterogeneity is large enough to overcome the separation between chapter removal counts.

3. Methods

Fig. 1 and Algorithm 1 organise the procedure around one external safety input and three economic decisions. The economic arm produces engineering-focus, scheduled-replacement, and incremental-stock evidence; the safety disposition determines the non-compensatory route (safety-led, dual, economic-led, routine, or safety-unassessed). A prospective hold-out then tests priority stability before register consolidation.

Note. The algorithm returns the ranking and pairwise inversions; it does not prescribe a universal top-k engineering-focus set. The five engineering-focus rows reported later are a case-study application.

3.1. Safety precedence and register architecture

COUR does not estimate airworthiness risk. Continuing-airworthiness safety risk depends on the probability and severity of consequences, followed by a tolerability decision and, where required, mitigation; SAE ARP5151A likewise provides a dedicated process for the ongoing safety assessment of aircraft and rotorcraft in commercial service [28,29]. Accordingly, each target enters the register with two distinct inputs: an operator-approved safety disposition and the economic evidence produced by this study. The rule is non-compensatory: any required safety action retains precedence, while COUR may inform resource planning but cannot relax that action (see Table 2).

ATA chapter, removal frequency, remark keywords, AOG cost, and COUR were not used as surrogates for safety criticality. Every case-study entry is classified as safety-unassessed pending an approved external disposition; the available data populate only the economic axis. The study consequently cannot test empirical concordance or divergence between safety and economic priority. This is a structural data boundary, not evidence of low or absent safety risk.

3.2. Fleet and observation window

The study fleet comprises 24 offshore medium twin-engine helicopters (OMTH) of a single type. The observation window spans 27 months from January 2024 to March 2026. Flight hours were

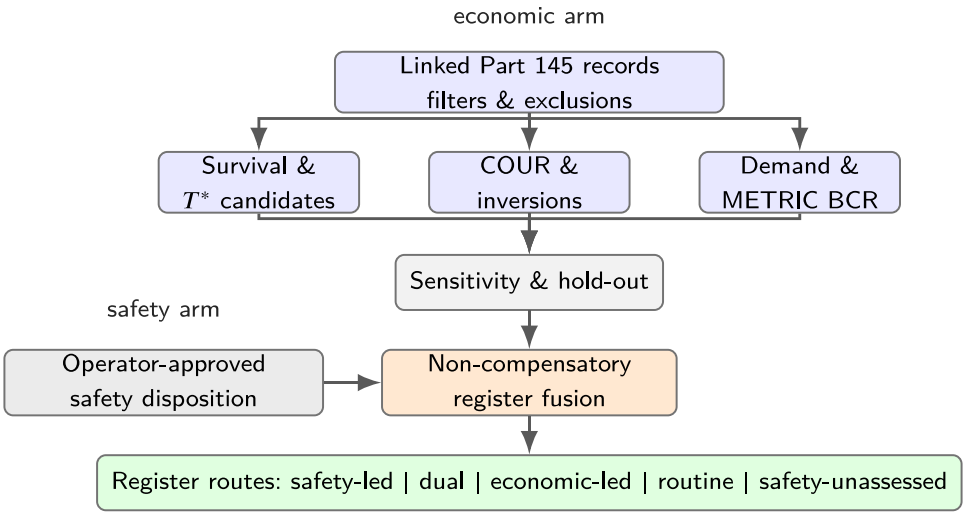


Fig. 1. Decision-flow architecture of the continuing-airworthiness review register. Economic evidence is produced in three streams and fused with an external safety disposition under a non-compensatory rule (Algorithm 1).

Table 2
Non-compensatory safety–economic decision rule for the continuing-airworthiness review register.

Operator-approved safety disposition	Economic screen	Register route	Decision rule
Safety action required	Any	Safety-led or dual	Retain the safety priority; COUR may inform resource planning but cannot relax the action.
No elevated safety action assigned	Eligible	Economic-led	Review on COUR, replacement, or stock evidence.
No elevated safety action assigned	Not eligible	Routine	No escalation from the evidence used in this study.
Not supplied	Any	Safety-unassessed	Obtain an approved safety disposition before an operational decision.

Note: The safety dispositions are workflow states supplied by the operator's approved process, not a new severity scale. ATA chapter, removal frequency, remark keywords, AOG cost, and COUR are not used as proxies for consequence severity or tolerability.

Algorithm 1 Construction of the continuing-airworthiness review register

Require: Operator-approved safety disposition; aircraft–month exposure; removals and replacement installations; work orders; repairable parameters; labour and AOG scenario values; export cutoff.
Ensure: Target; safety route/disposition; economic action type; evidence; uncertainty/scenario; proposed review action.
1: Link exposure, removals, work orders, and replacement installations; apply the discrepancy-driven removal definition and declared exclusions.
2: For each ATA chapter, compute parts, labour, and scenario-bounded AOG costs; form $\mathrm{COUR}_a = N_a u_a$; compare volume and COUR ranks and identify inversions by Eq. (3). Emit the engineering-focus ranking and AOG-scenario stability.
3: For each eligible component, chain the removed serial to the replacement serial; append right censoring when the replacement does not reappear as removed by the cutoff; apply delayed entry to pre-window installations.
4: Fit Weibull, log-normal, and exponential models to the same episodes; select by AIC and report BIC/KS as diagnostics. Proceed to age replacement only if the wear-out screen is met and the selected model yields a finite T^*. Emit the candidate, T^*, projected saving, and model qualification.
5: Estimate demand and dispersion; apply Poisson METRIC as baseline and NB as a stress test. If BCR > 1, emit an incremental-stock candidate; otherwise keep the item outside that set.
6: Attach prospective hold-out and sensitivity tests to the three economic outputs without converting them into safety evidence.
7: Merge economic outputs by target.
8: Apply the non-compensatory safety rule: required safety action with an economic candidate → dual (safety precedence); required safety action without an economic candidate → safety-led; no required safety action with an economic candidate → economic-led; neither → routine; unavailable safety disposition → safety-unassessed regardless of COUR.
9: **return** target, safety route/disposition, economic action type, evidence, uncertainty/scenario, and proposed review action. Never infer safety criticality from ATA, remarks, frequency, AOG cost, or COUR.

extracted from monthly utilisation logs, yielding 33,486.6 fleet hours (mean: 1395.3 h per aircraft and 1240.2 h per month). The rotorcraft type is omitted for confidentiality. Costs are reported as recorded.

3.3. Data sources

- *Flight-hour logs:* Monthly aircraft utilisation for exposure and demand rates.
- *Removal records:* Aircraft, ATA, part, installation-to-removal duration, removal date, and standard replacement cost, together with the coded reason, recorded condition, and pre-removal remark.
- *Aircraft log technical-log (ALTL) work orders:* Part 145 execution records linked to removals, providing ATA and man-hours.

Parts cost and linked labour are available in both temporal partitions, permitting COUR reconstruction by removal date. Every removal record also identifies the removed unit, its installation and removal dates, and the replacement component and its serial number; chaining these records within each component type reconstructs the install base without a separate roster export (Section 3.7). The exports also lack a reliable event-level AOG indicator, so grounding cost is a scenario bound rather than a measured rate (Section 3.5) (see Table 3).

3.4. Removal classification and exclusions

Only tracked rows coded as unscheduled discrepancies enter the candidate population; life-vest and life-jacket rows are then excluded because their recorded frequency is dominated by regulated inspection intervals rather than component condition. Work orders classified as aircraft-log history work-order (ALHWO) class (zero recorded man-hours) are identified separately. The 125 ALHWO documents (corresponding to 202 removals) are excluded from labour cost computation

Table 3
Minimum data schema for transferring the register workflow to another operator. Fields required for a complete approved safety assessment remain external to the economic model.

Entity	Minimum fields	Use
Aircraft–month exposure	Aircraft identifier, month, flight hours	Rates, NB panel, MCF, hold-out
Removal episode	Aircraft, component type/name, ATA, removed serial, installation date, removal date, standard cost, work-order key	Filters, COUR, survival, demand
Replacement installation	Removal-event key, replacement serial	Install-base reconstruction and right censoring
Work order	Key, ATA, corrective man-hours	COUR labour leg
Repairable parameters	Component, lead time, acquisition/investment cost	METRIC, EBO, BCR
Scenario parameters	Labour rate, AOG value, budget/fill-rate targets, export cutoff	COUR and stock sensitivities
Approved safety input	Target, disposition availability, required action/disposition	Non-compensatory register route

Table note. A complete safety assessment additionally requires failure condition, consequence severity, occurrence probability, approved risk/tolerability class, and mandatory operational disposition. Those fields are not inferred from ATA, remarks, frequency, AOG cost, or COUR.

but retained in removal counts and parts cost. Work orders spanning more than one ATA chapter (5.1% of the non-ALHWO total) are handled by proportional allocation of man-hours across the referenced chapters.

The recorded-reason audit first counts all mutually exclusive reason descriptions in the export, then applies the study filters sequentially so that the discrepancy universe, tracked subset, equipment exclusion, and final population remain reconcilable (Table 5).

Pre-removal remarks are normalised to uppercase ASCII and classified with transparent whole-word expressions, including recorded spelling variants. Seven physical categories (wear/erosion, corrosion/oxidation, play/clearance, damage/deformation, crack/fracture, leakage/fluid loss, and contamination/blockage) and a functional/electrical category are multi-label.

For compact reporting, the audit assigns each remark to one mutually exclusive primary class, in this priority order: physical evidence; functional/electrical evidence only; generic discrepancy/action wording; administrative/scheduled wording; other indeterminate text; or no useable remark. The union of the physical and functional classes defines specific anomaly evidence; its complement is a conservative bound on remarks without a specific anomaly, not an estimate of no-fault-found removals.

3.5. COUR decomposition

The cost of unreliability for ATA chapter a is:

$$\mathrm{COUR}_a = C_a^{\mathrm{parts}} + C_a^{\mathrm{labour}} + N_a \cdot d_{\mathrm{AOG}} \cdot r_{\mathrm{AOG}}, \tag{1}$$

where C_a^{parts} is the sum of standard parts replacement costs for all tracked removals in chapter a; $C_a^{\mathrm{labour}} = w_{\mathrm{labour}} \sum_{k \in a} H_k$ is the ATA-level labour cost at assumed hourly rate w_{labour} summed over the total man-hours H_k linked to chapter a; N_a is the chapter removal count; and $d_{\mathrm{AOG}} \cdot r_{\mathrm{AOG}}$ is the assumed AOG opportunity cost per removal event (duration in days times daily rate in USD).

Each removal contributes its full recorded standard replacement cost, irrespective of free-text action type; linked labour remains a separate leg (Section 4.2). Without an event-level grounding indicator (Section 3.3), the AOG term is applied uniformly as a scenario bound and may overstate removals completed on wing or during scheduled downtime.

For $N_a > 0$, let $q = d_{\mathrm{AOG}} r_{\mathrm{AOG}}$ denote the common scenario cost per event and define chapter-level COUR per removal as

$$u_a = \frac{C_a^{\mathrm{parts}} + C_a^{\mathrm{labour}}}{N_a} + q = \frac{\mathrm{COUR}_a}{N_a}, \qquad \mathrm{COUR}_a = N_a u_a. \tag{2}$$

For any two chapters with $N_a > N_b$, the lower-volume chapter b outranks a under COUR if and only if

$$\frac{u_b}{u_a} > \frac{N_a}{N_b}. \tag{3}$$

The observed volume and COUR weak orderings agree exactly under two conditions: (i) $u_b/u_a < N_a/N_b$ for every pair with $N_a > N_b$; and (ii) equal-count chapters have equal u_a. Equality in condition (i) creates a COUR tie that is absent from the volume ordering. Constant u_a across chapters is sufficient for agreement in one observed fleet but is not necessary: unequal per-removal costs may be too small to overcome the observed count gaps. It becomes necessary and sufficient only for agreement under every possible positive removal-count vector. To see necessity, if $u_b > u_a$, one can choose $1 < N_a/N_b < u_b/u_a$, which reverses the two rankings by Eq. (3).

Three AOG scenarios are defined (Section 4.4): Low ($15,000/day × 0.5 day = $7500 per removal), Mid ($20,000/day × 1.0 day = $20,000), and High ($25,000/day × 2.0 day = $50,000). The Mid scenario is adopted as the primary baseline. Labour rate sensitivity is assessed at $50, $100, and $150 per hour; the baseline is $50/h. For a continuous AOG cost q, write $B_a = C_a^{\mathrm{parts}} + C_a^{\mathrm{labour}}$, so $\mathrm{COUR}_a(q) = B_a + N_a q$. Each pair with unequal counts can change order only at $q^* = (B_b - B_a)/(N_a - N_b)$. The maximal non-negative interval containing the Mid value and excluding the nearest pairwise crossings is reported as the range that preserves every strict Mid-scenario COUR relation.

3.6. Removal rate and rank divergence

The ATA-level removal rate per 1000 flight hours is $r_a = 1000 \cdot N_a/E$, where $E = 33{,}486.6$ h. Volume rank is assigned by descending N_a; COUR rank by descending COUR_a. Both use competition ranks: tied values share the lowest occupied rank and the following rank retains the corresponding gap. Rank divergence is:

$$\Delta\mathrm{rk}_a = \mathrm{Vol.\ rank}_a - \mathrm{COUR\ rank}_a, \tag{4}$$

so positive values indicate chapters whose economic priority exceeds their removal-count priority.

To express the mechanism independently of the monetary scale, let $\bar{u} = \sum_a \mathrm{COUR}_a / \sum_a N_a$ be fleet mean COUR per removal and $z_a = \log(u_a/\bar{u})$. Since $\log(\mathrm{COUR}_a) = \log N_a + \log \bar{u} + z_a$, a lower-volume chapter b crosses chapter a exactly when $z_b - z_a > \log(N_a/N_b)$. Thus, per-removal cost dispersion creates rank divergence only when the pairwise cost deviation exceeds the corresponding count separation. We report the proportion of unequal-count chapter pairs satisfying this inequality as the pairwise inversion share; unlike an unqualified dispersion statistic, it maps directly to a change in economic ordering.

To assess whether Δrk values are sensitive to the particular removal sample, a nonparametric bootstrap uses 10,000 replicates. Each replicate resamples the 1711 events with replacement and recomputes parts cost and the AOG leg while holding ATA-level labour fixed. Percentile 95% confidence intervals (CI) are reported for the chapters with the largest observed $|\Delta\mathrm{rk}|$; a shift is treated as statistically robust when its interval excludes zero.

3.7. Wear-out screening

High-impact components (those appearing in the top Pareto layers by removal count or COUR contribution) are subjected to individual Weibull time-to-removal analysis. Time is measured in calendar days

from installation to unscheduled removal (days-on-aircraft). Throughout the survival analyses, $\delta = 1$ denotes an observed discrepancy-driven unscheduled removal; it is an analytical event indicator, not a claim of shop-confirmed functional failure. A two-parameter Weibull model is fitted by maximum likelihood:

$$F(t;\, \beta_w, \eta) = 1 - \exp\left[-\left(\frac{t}{\eta}\right)^{\beta_w}\right], \tag{5}$$

with shape parameter β_w and characteristic life η. Asymptotic 95% confidence intervals are reported for β_w. Components with $\beta_w > 1.05$ are classified as *wear-out*; $\beta_w < 0.95$ as *infant-mortality*; intermediate values as approximately random. The ± 0.05 margins are pragmatic indifference bands around the exponential boundary $\beta_w = 1$, not inferential cut-offs. Sensitivity therefore repeats point-estimate eligibility at $\beta_w > 1.00$ on the same top-20 component screen, while confidence intervals retain the uncertainty statement.

For each installed replacement, the reconstruction asks whether the same serial later appears as a removed unit. If it does, the later record closes the episode; if it does not, the episode remains right-censored at the export cutoff. Formally, within each component type p, every removed unit's installation episode is chained to the replacement unit installed at its removal. Let $i = 1, \ldots, m_p$ index the removal records for component type p. Record i identifies the removed-unit serial s_i, installation time τ_i, removal time υ_i, replacement component p_i^+, replacement serial s_i^+, and event indicator δ_i. Here, $\delta_i = 1$ when the record satisfies the study's discrepancy-driven unscheduled-removal definition and $\delta_i = 0$ for other withdrawals (e.g. scheduled overhaul). For a record whose replacement matches the same component type ($p_i^+ = p$) and whose replacement serial s_i^+ is observed, define the continuation indicator

$$\sigma_i = \begin{cases} 1, & \text{if } \exists\, i' \in \{1, \ldots, m_p\} : s_{i'} = s_i^+ \wedge \tau_{i'} \geq \upsilon_i, \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

Thus, $\sigma_i = 1$ identifies a subsequent removal record for the replacement serial. If no such record exists, the replacement component is treated as still installed at the export cutoff, and the procedure appends a right-censored episode:

$$\tau_{m_p+k} = \upsilon_i, \qquad \upsilon_{m_p+k} = t_{\text{cutoff}}, \qquad \delta_{m_p+k} = 0, \qquad t_{\text{cutoff}} = 31 \text{ Mar } 2026. \tag{7}$$

For the appended episode, observation begins when the replacement is installed and ends at the export cutoff. Setting $\delta_{m_p+k} = 0$ records that no subsequent removal was observed; it does not imply removal at the cutoff.

Reinstallations of the same physical unit (e.g. after repair, possibly on a different aircraft) generate a new episode index i rather than being deduplicated by serial number. The data set for component type p contains the m_p original removal episodes and one additional right-censored episode for every record that satisfies all three conditions: $\sigma_i = 0$, $p_i^+ = p$, and $s_i^+ \neq \varnothing$. Let $j = 1, \ldots, n_p$ index this full set. Duration and left-truncation entry age follow directly from the episode boundaries (τ_j, υ_j):

$$d_j = \upsilon_j - \tau_j, \qquad \ell_j = \max(0,\ t_0 - \tau_j), \qquad t_0 = 1 \text{ Jan } 2024, \tag{8}$$

where d_j is days-on-aircraft and ℓ_j is the left-truncation entry age (not to be confused with the ATA-chapter subscript a of Eq. (1) or the negative-binomial dispersion α of Section 3.10). For units installed before the extract window opened, ℓ_j conditions their likelihood contribution on survival to the window opening because their installation age at t_0 is known but their pre-window removal history is not observable [30]. Three specifications are compared: (a) uncensored, treating every removal as an observed event ($\delta_j \equiv 1$, no appended episodes); (b) right-censored using the reconstructed install base (Eqs. (6)–(7)); (c) right-censored with late entry (adding Eq. (8)). Specification (c) is used throughout the Results section unless stated otherwise; specification (a) is retained in sensitivity tables as the naive baseline. Table 4 reports, per component, the number of censored observations recovered ($n_p - m_p$), the fraction of removals for which no replacement serial is recorded (so no censored observation can be recovered), and the median late-entry age.

To assess distributional adequacy rather than claim model validation, three parametric survival models are compared for each wear-out candidate: Weibull (Eq. (5)), log-normal, and exponential, each fitted by maximum likelihood to the same right-censored, late-entry durations. Models are compared by the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC); absolute fit uses the Kolmogorov–Smirnov (KS) distance between fitted and empirical cumulative distribution functions (CDFs). AIC is the pre-specified selection criterion; BIC is a complexity-penalised diagnostic whose disagreements are reported without retrospective reselection. Because all parameters are estimated from the same data, KS p-values are biased low and are not reported; the statistic is used for within-part model comparison only. The model with the lowest AIC is selected as the preferred specification for that component (Table 11).

A residual limitation is not removable by this reconstruction: removals that occurred before 1 January 2024 are absent from the export entirely, so units removed before the window opened are never observed, even indirectly. Late entry corrects for units that survive into the window but were installed earlier; it does not recover units that did not survive that long. Pseudo-censor sensitivity (synthetic right-censored observations added in proportion to the observed event count, each censored at the mean days-on-aircraft for that part) was used in an earlier round as a stress test for the then-unavailable install base; it is retained only in the analytical artefacts as a historical cross-check, not as the basis for the reported estimates.

3.8. Cost-rate optimisation

For each wear-out component, the optimal scheduled replacement interval T^* minimises the long-run expected cost rate $C(T)$ under a block-replacement policy [2]:

$$C(T) = \frac{c_p + (c_u - c_p)\, F(T)}{\displaystyle\int_0^T [1 - F(t)]\, \mathrm{d}t}, \tag{9}$$

where c_p is the scheduled replacement cost (parts only, no AOG exposure assumed for a planned task), c_u is the unscheduled replacement cost (parts + ATA-level proportional labour + mid-scenario AOG cost), and $F(T)$ is the CDF of the fitted survival model at age T. The denominator is the expected time to replacement under the policy. Eq. (9) uses the AIC-selected model for each component (Table 11). Under Weibull, log-normal, or exponential survival, F and $1 - F$ take their corresponding closed forms, and numerical quadrature evaluates the denominator. Table 12 also reports a forced-Weibull result as a distributional-sensitivity comparator. The percentage saving at T^* is reported relative to purely reactive replacement ($T \to \infty$, cost rate $= c_u/\mu$, where μ is the fitted model's mean time to removal; $\mu = \eta\, \Gamma(1 + 1/\beta_w)$ under Weibull, $\mu = \exp(\mu_{\ln} + \sigma_{\ln}^2/2)$ under log-normal, and $\mu = \lambda$ under exponential).

Under a constant-hazard (exponential) fit, $C(T)$ decreases monotonically towards the reactive limit and has no interior minimum, so no finite T^* exists; the same can occur under log-normal when the hazard declines at long survival ages. Both are treated as findings, not missing values: T^* is reported as undefined and the component is excluded from scheduled-replacement candidacy under its selected model (Table 12), regardless of what a Weibull-forced fit would report.

Table 4
Install-base reconstruction audit: replacement units recorded at each removal and not subsequently observed as removals by 31 Mar 2026 are recovered as right-censored observations still installed at the export cutoff.

Part name	Observed removals	Recovered	% censored	Non-attrib.	Late entry med. (d)
Main wheel assy, 6800 kg	182	34	15.7	16	0
Upper scissor lever, pre-BT	107	22	17.1	9	0
Damper bolt	83	59	41.5	4	186
Upper rod end, post-BT	57	39	40.6	16	255
Tail-rotor scissor assy	57	42	42.4	4	362
Main wheel assy, 7000 kg	50	7	12.3	6	0
Main-rotor damper A	51	37	42.0	9	51
Damper rod-end assy	47	33	41.2	9	0
Lower scissor lever, pre-BT	39	18	31.6	3	71
Flight display unit A	34	9	20.9	29	44

Note: Recovered = replacement serials chained from a removal of the same component type that are not observed to be removed again by the export cutoff. Non-attributable = component removals for which the replacement component or serial is not recorded, so no censored observation can be recovered. Late entry: days between installation and the extract window start (1 Jan 2024) for units installed before the window opened; 0 for units installed within the window.

3.9. Part-level survival and subsystem recurrent-event analysis

Two estimands separate component age from fleet recurrent-removal burden. First, Nelson–Aalen cumulative hazards are fitted independently by component type, with the reconstructed install episodes supplying right-censored observations and delayed entry (Section 3.7). A component type is eligible only if it contributes at least ten observed removals. This ten-event minimum is a declared screening convention rather than an inferential criterion: it sets the smallest event count for which a component-level curve is reported, and its effect is stated rather than assumed (Section 4.8). These fits retain a coherent component-age risk set and are used only as component-level survival checks; hazards are not pooled across component types with different lives.

Second, subsystem burden is estimated as an aircraft-level mean cumulative function (MCF) for main rotor (ATA 62), landing gear (ATA 32), tail rotor (ATA 64), and rotor drive (ATA 63) recurrent removals. For subsystem g, $\widehat{M}_g(h) = \sum_{u \le h} d_g(u)/Y(u)$, where $d_g(u)$ is the number of recorded removals at cumulative in-window flight-hour exposure u and $Y(u)$ is the number of aircraft whose observed exposure reaches u [31].

Monthly removal counts are located at each aircraft's cumulative flight hours at month end, so multiple removals at the same exposure remain recurrent events. Aircraft without positive matched flight-hour exposure are excluded.

Pointwise 95% percentile intervals use 10,000 non-parametric bootstrap resamples of the 24 aircraft, with the same resample used across ATA curves. The exposure-supported primary horizon is 500 flight hours (21/24 aircraft at risk), 1000 h is secondary (15/24), and curves are truncated at 1500 h (12/24).

ATA 72 is excluded because its three removals fall below the declared ten-event screening minimum. The MCF estimates recorded removals per aircraft conditional on the observed window; it is neither a component hazard nor evidence of age-dependent degradation.

3.10. Stock-decision evidence: operational demand

Monthly flight logs report takeoff and engine-start counts alongside flight hours. An aircraft–ATA–month panel is constructed with removal count Y_{iam}, exposure H_{im} (flight hours), takeoffs per flight hour, and starts per flight hour. Poisson and negative-binomial (NB) generalised linear models are fitted with $\log(H_{im})$ offset and fixed effects for aircraft, ATA chapter, and calendar month [32]. Overdispersion (Poisson Pearson $\chi^2/\mathrm{df} > 1$) motivates the NB specification; the method-of-moments dispersion estimate α is the primary output used for METRIC calibration (Section 3.11). The usage covariates (takeoffs/FH and starts/FH) are exploratory: their joint incidence rate ratios are subject to collinearity and are not used as register inputs or operational scheduling levers.

3.11. METRIC benefit–cost stock screen

Spare-parts stocking recommendations are derived from a Poisson-demand single-echelon METRIC model [4]. For each part j, the expected pipeline (units in the repair–transport cycle) is $D_j = \lambda_j L_j$, where λ_j is the fleet-level demand rate (removals per day) estimated from the observation window and L_j is the assumed replenishment lead time (depot repair 30 days + transport 5 days = 35 days). The expected backorders (EBO) at stock level s are:

$$\mathrm{EBO}(s) = \sum_{k=s+1}^{\infty} (k - s)\, \frac{D_j^k\, e^{-D_j}}{k!}. \tag{10}$$

The benefit-cost ratio for the incremental stock level s_j recommended under Scenario S-B (minimum fill rate 85%) is:

$$\mathrm{BCR}_j = \frac{[\mathrm{EBO}_0 - \mathrm{EBO}(s_j)] \cdot r_{\mathrm{AOG}}}{s_j \cdot c_j}, \tag{11}$$

where EBO_0 is expected backorders at zero stock, r_{AOG} = \$20,000 day^{-1} is the mid AOG rate, and c_j is the unit part cost. Scenario S-B is the primary reporting baseline. Comparative scenarios (S-A: \$50k budget; S-C: minimum fill rate 95%; S-D: \$200k budget) are used only to confirm that the leading BCR ranks are preserved; detailed stock-level and scenario tables are retained in the analytical artefacts.

The EBO formula assumes Poisson demand arrivals, but the NB panel model (Section 3.10) yields a dispersion estimate $\alpha = 16.6$, indicating that removal counts at the aircraft–ATA level are substantially overdispersed relative to Poisson. Under a negative-binomial demand model, variance in pipeline fill exceeds the Poisson prediction for the same mean, so the Poisson EBO values are optimistic for high-dispersion parts. Across the budget and fill-rate alternatives above, the ranking of leading BCR candidates is preserved, indicating that the Poisson baseline is adequate for the triage purpose of the BCR screen even under the observed overdispersion.

3.12. Priority stability: prospective hold-out

Priority stability is assessed in two parallel tests using a 21-month development period (January 2024–September 2025) and a six-month hold-out (October 2025–March 2026).

Removal layer. Development-period rates are multiplied by hold-out exposure, and Spearman ρ compares forecast and observed ranks across the 27 chapters with non-zero development activity.

COUR layer. Parts, labour, and Mid-scenario AOG costs are reconstructed in each period. The predictive COUR forecast multiplies development-period COUR per 1000 flight hours by hold-out exposure; Spearman ρ and Kendall τ assess rank and level agreement. With the pipeline fully specified, Section 4 reports its application to the 27-month fleet record.

Table 5
Recorded-reason and pre-removal remark audit.

Audit block	Recorded disposition or evidence class	*n*	%
Coded-reason universe	Unscheduled removal/discrepancy	4422	60.5
	Overhaul	1491	20.4
	Repair-station inspection	831	11.4
	Supply-related removal	358	4.9
	Troubleshooting	115	1.6
	Retirement	90	1.2
	Component swap	1	0.0
Selection	Tracked discrepancy rows before equipment exclusion	2065	–
	Life-vest/life-jacket rows excluded	354	–
	Final analytical population	1711	100.0
	Recorded as unserviceable	1711	100.0
	Useable pre-removal remark	1707	99.8
Primary remark class	Physical-anomaly evidence	937	54.8
	Functional/electrical evidence only	442	25.8
	Generic discrepancy/action only	178	10.4
	Administrative/scheduled wording only	82	4.8
	Other indeterminate free text	68	4.0
	No useable remark	4	0.2
Independent check	Explicit no-fault phrase	0	0.0
	No specific anomaly in remark (conservative bound)	332	19.4

Notes: Percentages in the coded-reason block use the 7308 rows with a populated reason; primary remark classes are mutually exclusive and use the 1711-event study population. Physical-anomaly evidence includes wear/erosion, corrosion/oxidation, play/clearance, damage/deformation, crack/fracture, leakage/fluid loss, or contamination/blockage. Lexical categories describe the pre-removal wording and may not be interpreted as teardown-confirmed shop findings. The no-fault check used whole-word expressions including NFF, no fault found, no defect, nil defect, no trouble found, tested OK, and serviceable; the last term does not match unserviceable.

4. Results

Results follow the three decisions delivered by the airworthiness review register (Algorithm 1). The engineering-focus evidence comes first (Sections 4.1–4.4): removal burden, labour, COUR, and robustness. Survival and cost-rate results for scheduled replacement follow in Sections 4.5–4.7, with aircraft-level recurrent burden in Section 4.8. Sections 4.9–4.10 then supply the stock evidence. Prospective tests then assess whether the dominant priorities persist before Section 4.13 consolidates the evidence streams.

4.1. Engineering-focus evidence: removal burden

Over the 27-month window, 1711 tracked unscheduled mechanical removals were recorded across 27 ATA chapters. Table 6 lists the ten highest-volume chapters. ATA 62 (main rotors) leads with 541 removals at 16.16 per 1000 flight hours, followed by ATA 32 (landing gear, 282 removals, 8.42 per 1000 h) and ATA 31 (indicating and recording, 148 removals, 4.42 per 1000 h). These three chapters together account for 56.8% of all unscheduled removals; the top ten cover 87.7%.

Table 5 reconciles this population. Of 7308 export rows with a coded reason, 4422 are coded as unscheduled discrepancies; 2065 of these are tracked, and excluding 354 life-vest/life-jacket rows yields the 1711-event analytical population. All 1711 are recorded as unserviceable, and 1707 contain a useable pre-removal remark. The conservative primary classification finds physical-anomaly evidence in 937 records and functional/electrical evidence only in 442, so 1379 (80.6%) contain a specific anomaly. The remaining 332 (19.4%) contain only generic discrepancy/action wording (178), administrative/scheduled wording (82), other indeterminate text (68), or no useable remark (4). No whole-word no-fault marker was found. These administrative and pre-removal fields support the term *discrepancy-driven removal*; they do not establish a shop-confirmed defect.

4.2. Corrective maintenance burden

Table 7 decomposes corrective maintenance burden for the ten highest work-order-volume ATA chapters. Fleet-wide, 1200 non-ALHWO work orders consumed 33,565 man-hours. ATA 62 accounts for the largest labour mass (13,905 man-hours; 29.7 h per removal), reflecting the complexity of main rotor rectification. ATA 64 (tail rotor) records 54.5 h per removal, the highest per-removal labour intensity among high-count chapters, whereas ATA 31 requires only 9.2 h per removal. ATA 29 (hydraulic power) has the highest ratio (109.7 h/removal), driven mainly by one proportionally allocated multi-task inspection document; this is document-level labour, not hours applied uniformly to each ATA 29 event. These contrasts in labour intensity are not reflected in removal count alone and directly feed the labour leg of the COUR decomposition in the next subsection.

Table 6
Top ATA systems by tracked unscheduled mechanical removal count (R-1).

Rank	ATA	Removals	Rate (per 1000 h)	Cum. share (%)
1	62 (Rotor(s))	541	16.16	31.6
2	32 (Landing gear)	282	8.42	48.1
3	31 (Indicating/recording)	148	4.42	56.8
4	64 (Tail rotor)	91	2.72	62.1
5	21 (Air conditioning)	82	2.45	66.9
6	25 (Equipment/furnishings)	82	2.45	71.7
7	63 (Rotor drive)	74	2.21	76.0
8	22 (Auto flight)	73	2.18	80.2
9	24 (Electrical power)	71	2.12	84.4
10	34 (Navigation)	57	1.70	87.7

Note: Rate normalised by total fleet exposure (33,486.6 flight hours). Life-vest and other regulated safety-equipment removals are excluded.

4.3. Cost of unreliability

Table 8 presents the COUR decomposition by ATA chapter under the mid AOG scenario. Research question 1 is answered on an economic proxy: scenario-bounded COUR and per-event parts cost rank review priority; safety outcomes and regulatory risk scores are not measured here. Fleet-wide COUR totals $92.21M: parts $56.31M (61.1%), AOG opportunity cost $34.22M (37.1%), and labour $1.68M (1.8%). Fig. 2 shows the cumulative COUR Pareto.

Labour is computed from ATA-level linked work-order man-hours at the assumed rate of $50/h, excluding ALHWO work orders with zero recorded man-hours. The AOG leg uses the mid bounding scenario:

Table 7
Corrective maintenance burden by ATA system. Man-hours are proportionally allocated to ATA chapters for the 5.1% of work orders spanning more than one ATA system. ALHWO work orders carry zero recorded man-hours and are counted separately.

Rank	ATA (system)	WOs	ALHWO	Removals	Man-h	h/rem.	NST
1	62 (Rotors)	230	40	468	13,905	29.7	815
2	64 (Tail rotor)	33	11	63	3431	54.5	172
3	32 (Landing gear)	268	3	276	3354	12.2	476
4	34 (Navigation)	52	3	53	2086	39.4	68
5	63 (Rotor drive)	58	8	63	1677	26.6	160
6	29 (Hydraulic power)	12	5	13	1426	109.7	37
7	31 (Indicating/recording)	134	10	136	1254	9.2	187
8	21 (Air conditioning)	78	2	80	1156	14.5	156
9	25 (Equipment/furnishings)	52	27	54	1015	18.8	78
10	22 (Auto flight)	68	3	69	979	14.2	153
Fleet total (all ATA)		1200	125	–	33,565	–	–

Notes: WOs = non-ALHWO ALTL execution documents linked to tracked unscheduled mechanical removals. ALHWO = work-order class with zero recorded man-hours; total 125 ALHWO documents (202 removals) are excluded from man-hour totals. Man-h = total man-hours; h/rem. = man-hours per removal; NST = non-scheduled tasks allocated from linked work orders.

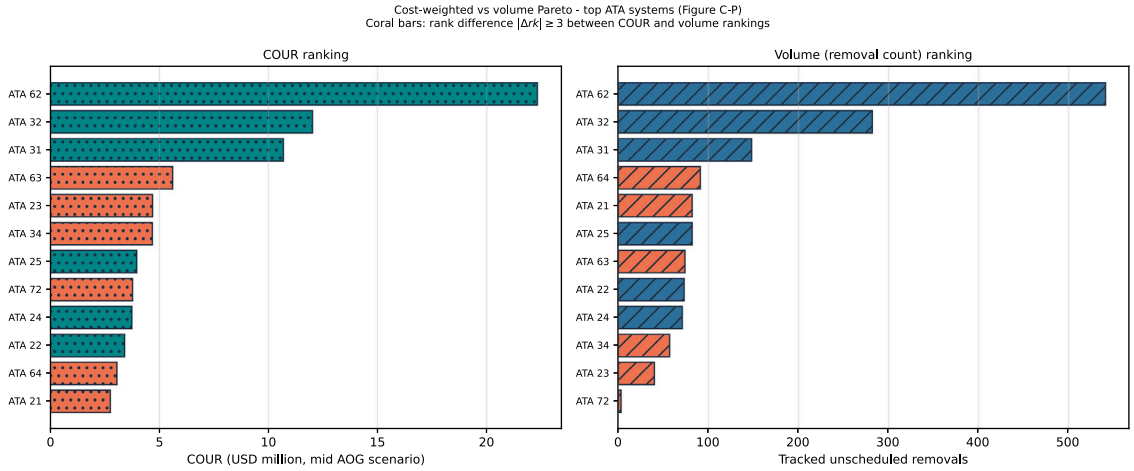


Fig. 2. Cumulative Pareto of fleet cost of unreliability by ATA chapter (mid scenario). The three leading chapters (ATA 62, 32, 31) account for 48.8% of fleet COUR and anchor the engineering-focus rows of the airworthiness review register (Table 15).

Table 8
Cost of unreliability (COUR) by ATA system, Mid AOG scenario. Labour cost: $50/h assumed rate. AOG cost: 1.0 day × $20,000/day per unscheduled removal event (stated as a bounding assumption; not all removals ground the aircraft). Volume rank (Vol. rk) and COUR rank divergence (Δrk) highlight systems where economic priority differs from removal frequency. COUR/rem. is the chapter-level per-removal burden u_a in Eq. (2).

Rk	ATA (system)	Removals	Parts ($k)	Labour ($k)	AOG ($k)	COUR ($k)	COUR/rem. ($k)	Vol. rk	Δrk
1	62 (Main rotors)	541	10 805.2	695.3	10 820.0	22 320.5	41.3	1	0
2	32 (Landing gear)	282	6 208.6	167.7	5 640.0	12 016.3	42.6	2	0
3	31 (Indicating/recording)	148	7 660.0	62.7	2 960.0	10 682.7	72.2	3	0
4	63 (Rotor drive)	74	4 047.1	83.8	1 480.0	5 610.9	75.8	7	+3
5	23 (Communications)	40	3 877.0	12.6	800.0	4 689.6	117.2	11	+6
6	34 (Navigation)	57	3 436.4	104.3	1 140.0	4 680.7	82.1	10	+4
7	25 (Equipment/furnishings)	82	2 274.0	50.8	1 640.0	3964.7	48.4	5	−2
8	72 (Engine)	3	3 710.2	3.9	60.0	3 774.1	1258.0	25	+17
9	24 (Electrical power)	71	2 290.9	30.1	1 420.0	3 741.0	52.7	9	0
10	22 (Auto flight)	73	1 897.9	49.0	1 460.0	3 406.9	46.7	8	−2
11	64 (Tail rotor)	91	1 066.8	171.5	1 820.0	3 058.4	33.6	4	−7
12	21 (Air conditioning)	82	1 056.3	57.8	1 640.0	2 754.1	33.6	5	−7
Fleet total (all ATA)		1711	56 313.2	1678.2	34 220.0	92 211.4	53.9	–	–

Notes: All costs in United States dollars (USD) thousands. Parts cost = sum of standard replacement cost from removal records. Labour cost = ATA-level man-hours from linked work orders × assumed rate. ALHWO work orders (zero man-hours) excluded from labour. COUR/rem. is total COUR divided by removal count. Tied values receive competition ranks (1, 1, 3, …). Δrk = volume rank − COUR rank; positive values indicate systems whose cost priority exceeds their removal-count priority.

1.0 day × $20,000/day per unscheduled removal event. This is an assumption-bound screening convention (Section 3.3): the operator export contains no event-level grounding indicator, so the AOG leg must not be read as a measured downtime cost. The rank-divergence column reports Δrk = volume rank − COUR rank, so positive values mark systems whose economic priority exceeds their removal-count priority.

The Mid-scenario fleet mean is $\bar{u}$ = $53.9k per removal. Chapter values range from $20.9k to $1.258M per removal, a 60.1-fold span, with an event-weighted coefficient of variation of 1.04. Of 346 chapter pairs with unequal removal counts, 64 (18.5%) satisfy Eq. (3); cross-sectional concordance is consequently incomplete (Spearman $\rho = 0.801$, Kendall $\tau = 0.626$). The result also explains why heterogeneous costs need not disrupt every priority: none of the per-removal cost gaps overturns the

Table 9
Bootstrap robustness of ATA rank divergence (Δrk) under event-level resampling. Each of 10,000 replicates draws 1711 removal records with replacement; ATA-level labour cost is held fixed at observed work-order totals and parts and AOG legs are recomputed from the resampled events. Ties receive competition ranks.

ATA	System	Obs. Δrk	Vol. rk	COUR rk	Boot. mean	95% CI	Robust?
72	Engine	+17	25	8	+14.8	[0, 20]	No
23	Communications	+6	11	5	+5.0	[0, 7]	No
63	Rotor drive	+3	7	4	+1.5	[−5, 5]	No
64	Tail rotor	−7	4	11	−5.7	[−8, −3]	Yes
21	Air conditioning	−7	5	12	−5.6	[−8, −3]	Yes

Notes: Δrk = volume rank − COUR rank. Volume and COUR ranks use the competition convention: tied values share the lowest occupied rank and the next rank retains the gap. "Robust?" indicates whether the percentile bootstrap 95% confidence interval excludes zero. Mid AOG scenario; labour at $50/h.

count separation among the three highest-volume chapters. Conversely, ATA 72 has only three removals but u_{72} = $1.258M, 23.3 times the fleet mean, and crosses 17 higher-volume chapters.

The top three ATA chapters by COUR are the same as by volume (ATA 62: $22.32M; ATA 32: $12.02M; ATA 31: $10.68M), but their relative magnitudes and the ordering of remaining chapters differ substantially. The most pronounced rank divergences are:

- ATA 72 (engine): Δrk = +17 (volume rank 25, COUR rank 8; $3.77M driven by high per-event parts cost with only 3 removals).
- ATA 23 (communications): Δrk = +6 (volume rank 11, COUR rank 5; $3.88M in parts cost across 40 removals, $97k average per removal).
- ATA 63 (rotor drive): Δrk = +3 (volume rank 7, COUR rank 4).
- ATA 64 (tail rotor): Δrk = −7 (volume rank 4, COUR rank 11; high removal count but relatively modest parts cost per event).
- ATA 21 (air conditioning): Δrk = −7 (competition volume rank 5, tied with ATA 25).

These divergences confirm that volume-based triage would under-prioritise high-parts-cost, low-frequency systems and over-prioritise high-count, low-cost systems. Bootstrap confidence intervals (Table 9) show that downward shifts for ATA 64 and ATA 21 are statistically robust (95% CI $[-8, -3]$ for each, Δrk < 0 in all 10,000 replicates). The ATA 72 upward shift is directionally stable (95.2% of replicates positive; median +16), but its 95% CI $[0, 20]$ touches zero because three observed removals can disappear under resampling. ATA 72 therefore enters the action register as a *hidden-cost screening candidate*, with caution on the exact rank gap.

4.4. COUR sensitivity analysis

This subsection tests whether the engineering-focus priorities survive the principal cost assumptions before the analysis moves from engineering focus to prescriptive actions. Detailed Low/Mid/High and parts–labour decompositions remain in the analytical artefacts; the checks below summarise the register-relevant outcomes.

AOG Low/Mid/High.
Fleet COUR is $70.82M/$92.21M/$143.54M; the parts leg remains fixed at $56.31M and the top-three share ranges from 46.4% to 51.7%. The leading set ATA 62/32/31 is preserved, although Low orders it as 62–31–32 whereas Mid and High order it as 62–32–31.

Exact Mid-order band.
The open interval 17,586.51 < q < 20,120.72 USD per removal preserves every strict Mid-scenario COUR relation. The lower boundary is the ATA 25–ATA 72 crossing; the upper boundary is the ATA 45–ATA 33 crossing (Mid ranks 18 and 19). ATA 72 remains a hidden-cost focus row in all three scenarios, but the fifth focus row is scenario-dependent (ATA 45 under Low, ATA 23 under Mid, ATA 34 under High).

AOG removed (parts + labour).
Setting AOG to zero does not hide any high-parts chapter: the top-12 ranks remain economically ordered, ATA 72 rises to parts–labour rank 6 ($\Delta\text{rk}_{\text{PL}}$ = +19), and Mid-COUR ranks agree closely with the parts–labour order (Spearman $\rho = 0.97$).

Labour rate $50/$100/$150 h^{-1}.
Fleet COUR rises from $92.21M to $93.89M and $95.57M; labour remains at most 5.3% of COUR. The top-three ranking (ATA 62, 32, 31) is unchanged.

ALHWO imputation.
Imputing ALHWO labour at each chapter's non-ALHWO median man-hours per removal adds 5,893 h across 202 ALHWO-linked removals (+$0.30M labour). Under Mid AOG and $50/h, fleet COUR rises from $92.21M to $92.51M; parts, AOG, the leading-chapter ranking, and the top-three COUR share (48.8%) are unchanged.

Normalised fleet metrics. Research question 4 calls for exposure-normalised burden indicators. Tables 6 and 8 provide the ATA-level basis; the following fleet indicators support cross-fleet comparison. Over 33,486.6 flight hours and 27 months on 24 aircraft (54 aircraft-years), mid-scenario COUR averages $2.75 million per 1000 fleet flight hours and $1.71 million per aircraft-year. Tracked unscheduled removals average 51.1 events per 1000 flight hours and 31.7 events per aircraft-year. Mean fleet exposure is 1395.3 flight hours per aircraft over the window (620.1 flight hours per aircraft-year). ATA 62 alone accounts for $0.67 million COUR per 1000 flight hours (Table 8), showing chapter burden relative to observed fleet exposure.

4.5. Wear-out screening

With economically and volumetrically dominant ATA chapters identified, the prescriptive leg begins with wear-out screening on high-impact components drawn from the Pareto layers above. Nine of ten high-impact components exhibit wear-out shape (β_w > 1.05) under right-censored late-entry fits from the reconstructed install base (specification (c); Tables 4 and 10). The single exception is the flight display unit (ATA 31, $\beta_w = 0.94$, infant-mortality class), which is not advanced to age-replacement screening. Its shape parameter rises from 0.73 under the uncensored fit to 0.94 under (c) — closer to, but still below, the 0.95 boundary used to classify approximately random hazard. Among wear-out components, shape parameters range from $\beta_w = 1.51$ (main wheel, 6800 kg variant) to $\beta_w = 2.49$ (rod end assembly, damper), indicating a spectrum from mild to pronounced wear-driven degradation. Characteristic lives span 84 days (main wheel, 7000 kg variant) to 755 days (scissor assembly, tail rotor). The 95% confidence intervals on β_w exclude 1.0 for all nine wear-out components under specification (c), a stronger evidentiary base for the wear-out label than the uncensored fit (seven of nine).

For Flight display unit A, the specification-(c) estimate is $\beta_w = 0.941$ (95% CI 0.703–1.179; n = 59, including 25 censored observations; $\eta = 465.3$ days). Because the interval spans the exponential boundary,

Table 10
Weibull time-to-removal parameters for high-impact components, right-censored with late entry from the reconstructed install base (specification (c); Table 4).

Part name	n	Cens.	β_w	β_w 95% CI	η (d)	Med. (d)	Class
Main wheel assy, 6800 kg	271	89	1.51	[1.35, 1.66]	159	108	Wear-out
Upper scissor lever, pre-BT	130	23	1.61	[1.39, 1.83]	191	115	Wear-out
Damper bolt	142	59	1.76	[1.48, 2.04]	668	576	Wear-out
Upper rod end, post-BT	97	40	1.59	[1.31, 1.88]	720	452	Wear-out
Tail-rotor scissor assy	99	42	1.76	[1.47, 2.04]	755	610	Wear-out
Main wheel assy, 7000 kg	62	12	1.65	[1.32, 1.97]	84	49	Wear-out
Main-rotor damper A	93	42	2.05	[1.59, 2.52]	657	462	Wear-out
Damper rod-end assy	81	34	2.49	[1.94, 3.04]	544	346	Wear-out
Lower scissor lever, pre-BT	57	18	1.64	[1.24, 2.04]	463	379	Wear-out
Flight display unit A	59	25	0.94	[0.70, 1.18]	465	447	Infant

Note: CI = confidence interval. Time scale: days on aircraft since installation. Under specification (c), replacement units not subsequently observed as removals by 31 Mar 2026 are recovered as right-censored observations (Cens. column), and units installed before the 1 Jan 2024 extract window enter the likelihood at their observed age. n = observed removals + recovered censored observations. Median (Med.) is the median observed time-to-removal, unaffected by the censoring specification. 9 of 10 components exhibit wear-out shape ($\beta_w > 1.05$), candidates for age-replacement analysis; none change hazard-shape class relative to the uncensored fit.

"infant mortality" is a point-estimate flag for possible early-life removal concentration, not a confirmed failure mechanism. The appropriate follow-up is to inspect installation, repair, and early-removal records; these data do not support an age-replacement recommendation for this component.

On the fixed top-20 screen, lowering the point threshold from $\beta_w > 1.05$ to $\beta_w > 1.00$ adds no component: 13 of 20 qualify under the uncensored specification at either threshold, and 16 of 20 under specification (c) at either threshold. Applying (c) at the screening stage would add the condenser assembly, lower scissor lever, post-BT, and smart linear actuator to the original 13-component point-estimate set, but Table 11 retains that original population to avoid retrospective reselection. Confidence intervals exclude 1 for 9 of the 13 uncensored candidates and 10 of the 16 specification-(c) candidates, underscoring that the point threshold is a screen rather than an inferential finding.

Table 4 shows that 12%–42% of removals per component recover a right-censored replacement still installed at the export cutoff, with a further 4.6%–21.6% (excluding the infant-mortality contrast, 36.7%) unattributable because no replacement serial is recorded. Late entry is material for six of ten components (median 51–362 days), confirming that a non-trivial fraction of the observed population was already installed before the extract window opened. Legacy pseudo-censor stress tests from an earlier round, when the install base was assumed unavailable, are retained only in the analytical artefacts and no longer inform the main-text parameters.

4.6. Distributional assessment for scheduled replacement

Once wear-out shape and characteristic life are established, distributional adequacy is checked before age-replacement optimisation. AIC selects Weibull for 7 of 13 wear-out candidates from the top-20 removal screen, log-normal for 5, and exponential for 1 under specification (c) (Table 11). BIC agrees with AIC for 11 of 13 components; for the HEEL battery and main-rotor blade, AIC selects Weibull whereas BIC selects exponential, with small criterion differences consistent with model uncertainty rather than a robust distributional distinction. Relative to the uncensored screen, four components change AIC-selected model. Two are clear shifts (ΔAIC > 10): the upper rod end, post-BT moves from Weibull to log-normal, and the lower scissor lever, pre-BT (β_w = 1.64 under (c)) moves from exponential to Weibull, reinstating it as an age-replacement candidate (Section 4.7). Two are near-ties on both sides (ΔAIC < 1.5) and are not read as robust reversals: the main-rotor blade (β_w = 1.24 under (c)) and the Marathon 44-Ah battery both sit at the exponential–Weibull indifference point under either specification. For the leading prescriptive candidates, namely the damper bolt and damper rod-end assembly, Weibull and log-normal both outperform exponential by wide margins (ΔAIC > 20), confirming that a time-varying hazard is appropriate for these parts and that age-replacement analysis is well-supported by the distributional evidence.

4.7. Age-replacement candidates

Nine of the twelve wear-out candidates retain a finite cost-rate minimum under the AIC-selected model at specification (c) (Table 12). A forced-Weibull result at the same specification provides the distributional-sensitivity comparator. Selected-model savings range from 68.2% (damper bolt, Weibull) to 0.8% (main-rotor damper B, Weibull). Three components — both main wheel variants and the main-rotor blade — have no finite T^* under their selected model: the cost rate decreases monotonically towards the reactive limit rather than reaching an interior minimum, so scheduled age replacement is not economically justified for these under the stated cost assumptions. In this table, c_p is the scheduled replacement cost using parts only, while c_u combines parts, ATA-level labour, and the mid AOG assumption of $20,000 per event. Savings measure the reduction in expected cost rate relative to purely reactive replacement, and the rows are sorted by savings in descending order.

Fig. 3 plots the four largest-savings cost-rate functions under their selected model. Vertical markers identify the cost-rate-minimising age T^*, and the damper bolt exhibits the deepest minimum, consistent with its 68.2% saving and high c_u/c_p ratio ($21,702 versus $417).

The damper bolt (Weibull, 68.2% saving) and damper rod-end assembly (log-normal, 49.8%) are the strongest candidates, driven by a favourable ratio of scheduled cost (c_p = $417 and $4079, respectively) to unscheduled cost (c_u = $21,702 and $25,364). The HEEL battery (Weibull, 43.0%), tail-rotor scissor assembly (Weibull, 41.4%), upper scissor lever, pre-BT (log-normal, 41.1%), and upper rod end, post-BT (log-normal, 34.2%) offer comparable moderate-to-strong savings once the AIC-selected model replaces a Weibull assumption forced on components the distributional screen does not support. The lower scissor lever, pre-BT (Weibull, 23.1%) is a corrected candidate: the uncensored screen (Table 11) preferred an exponential fit for this component, under which no finite T^* exists, so its earlier inclusion at forced-Weibull T^* was internally inconsistent; under (c) Weibull is itself AIC-selected and the candidacy is now well founded. The remaining candidates with a finite T^* (both main-rotor damper variants) yield savings below 2%, where the cost-rate minimum is shallow and prescriptive scheduling is sensitive to parameter uncertainty.

Without the $20,000 AOG leg, the damper-bolt saving falls sharply and T^* lengthens under the retired uncensored baseline; removing labour as well eliminates a finite T^* when $c_u/c_p \approx 1$. Engineering teams with verified AOG duration should substitute that value before committing to a review interval. Legacy pseudo-censor and c_u-composition tables relative to the superseded uncensored baseline are retained only in the analytical artefacts.

Table 11
Survival model comparison for wear-out components, right-censored with late entry from the reconstructed install base (specification (c); Table 4). AIC is the pre-specified selection criterion; BIC is reported as a diagnostic.

Part name	n	Cens.	β_w	AIC			BIC			KS distance			Best AIC	Best BIC
				W	LN	Exp	W	LN	Exp	W	LN	Exp		
Main wheel assy, 6800 kg	271	89	1.51	2156	**2127**	2201	2163	**2134**	2205	0.189	**0.138**	0.178	LN	LN
Upper scissor lever, pre-BT	130	23	1.61	1283	**1253**	1316	1289	**1258**	1319	0.170	**0.155**	0.272	LN	LN
Damper bolt	142	59	1.76	**1197**	1208	1231	**1203**	1214	1234	**0.080**	0.102	0.188	W	W
Upper rod end, post-BT	97	40	1.59	839	**828**	854	844	**833**	856	0.198	**0.183**	0.253	LN	LN
Tail-rotor scissor assy	99	42	1.76	**824**	880	852	**830**	886	855	**0.141**	0.182	0.294	W	W
Main wheel assy, 7000 kg	62	12	1.65	522	**513**	539	527	**517**	541	0.221	**0.172**	0.200	LN	LN
Main-rotor damper A	93	42	2.05	**739**	740	766	**744**	746	769	**0.162**	0.129	0.207	W	W
Damper rod-end assy	81	34	2.49	649	**647**	690	653	**651**	692	0.239	**0.196**	0.267	LN	LN
Lower scissor lever, pre-BT	57	18	1.64	**536**	553	547	**540**	557	549	**0.087**	0.156	0.214	W	W
HEEL battery	215	186	1.33	**514**	515	515	521	522	**518**	**0.676**	0.683	0.689	W	Exp
Main-rotor blade	44	18	1.24	**385**	417	386	389	421	**387**	**0.477**	0.505	0.510	W	Exp
Main-rotor damper B	35	17	1.89	**263**	267	265	**266**	270	266	**0.192**	0.283	0.450	W	W
Marathon 44-Ah battery	74	58	1.19	258	260	**256**	263	265	**259**	0.760	0.756	**0.755**	Exp	Exp

Note: Models are Weibull (W), log-normal (LN), and exponential (Exp), right-censored with late entry (specification (c)). Only components with Weibull wear-out shape under the uncensored screen ($\beta_w > 1.05$, $n \geq 8$ observed removals) are included; this population remains fixed so specifications are comparable. AIC remains the pre-specified selection criterion; BIC is a complexity-penalised diagnostic and does not trigger retrospective model reselection. AIC and BIC disagree for 2 of 13 components: HEEL battery (AIC: W; BIC: Exp); Main-rotor blade (AIC: W; BIC: Exp). For heavily censored components, KS distance against observed removal times alone is a secondary diagnostic. AIC selects W for 7 of 13 components, LN for 5, and Exp for 1.

Table 12
Age-replacement optimisation under the AIC-selected model per component (specification (c); Table 11), compared with T^* forced under Weibull at the same specification. Components without a finite T^* under their selected model are not scheduled-replacement candidates under the stated cost assumptions.

Part name	Model	c_p ($)	c_u ($)	T^*_{sel} (d)	$Save_{sel}$ (%)	T^*_W (d)	$Save_W$ (%)
Damper bolt	Weibull	417	21,702	84	68.2	84	68.2
Damper rod-end assy	LogNormal	4,079	25,364	205	49.8	240	45.0
HEEL battery	Weibull	327	20,667	192	43.0	192	43.0
Tail-rotor scissor assy	Weibull	2,042	23,927	232	41.4	232	41.4
Upper scissor lever, pre-BT	LogNormal	3,640	24,925	59	41.1	90	22.3
Upper rod end, post-BT	LogNormal	5,093	26,378	279	34.2	428	15.3
Lower scissor lever, pre-BT	Weibull	3,729	25,014	217	23.1	217	23.1
Main-rotor damper A	Weibull	29,613	50,898	840	1.5	840	1.5
Main-rotor damper B	Weibull	29,613	50,898	1169	0.8	1169	0.8
Main wheel assy, 6800 kg	LogNormal	17,003	37,598	–	–	283	0.5
Main wheel assy, 7000 kg	LogNormal	19,153	39,748	–	–	125	1.0
Main-rotor blade	Weibull	158,970	180,255	–	–	–	–

Note: Costs are in USD; replacement ages are in days. T^*_{sel}/$Save_{sel}$: optimal age and cost-rate saving under the model AIC-selects for that component at specification (c) (Table 11). T^*_W/$Save_W$: same optimisation forced under Weibull, same specification, reported as a distributional sensitivity check. Three components (both main wheels, main rotor blade) have no finite T^* under their selected model: the cost rate decreases monotonically towards the reactive-replacement limit rather than reaching an interior minimum, so scheduled age replacement is not economically justified for these under the stated cost assumptions — for the log-normal-selected main wheels this reflects a declining hazard tail at long survival ages, not a constant-hazard assumption. Two components (HEEL battery, second main rotor damper) were not in the ten-component screen of Table 10 but are included here from the wider 13-component wear-out population of Table 11. Ranked by $Save_{sel}$.

4.8. Subsystem recurrent-removal burden

The component-level screen yields 19 eligible Nelson–Aalen fits: 11 main-rotor, three landing-gear, three rotor-drive, and two tail-rotor component types. The cost of the ten-event screening minimum (Section 3.9) is therefore small and quantified: the eligible fits still cover 87.6%, 86.9%, 81.1%, and 81.3% of removals in ATA 62, 32, 63, and 64 respectively, while ATA 72, with three removals, is left without a component-level curve. Each fit incorporates reconstructed right-censoring and delayed entry rather than treating every removal in a heterogeneous ATA chapter as a complete lifetime.

The recurrent-event analysis uses 952 of the 988 removals in the four eligible chapters. Thirty main-rotor and six landing-gear removals are excluded because their aircraft has no matched flight-hour history; four retained main-rotor removals occur before positive in-window exposure accumulates and form a tied jump at zero. At the pre-specified 500-hour horizon, the MCF is 8.00 recorded removals per aircraft for ATA 62 (95% CI 6.17–9.91), 4.36 for ATA 32 (3.45–5.51), 1.56 for ATA 64 (0.81–2.37), and 1.32 for ATA 63 (0.74–2.04). The ATA 62–ATA 32 difference is 3.64 removals per aircraft (1.93–5.43), whereas the ATA 64–ATA 63 difference is only 0.24 (−0.78–1.26). At 1000 h, the corresponding MCFs are 14.87 (11.81–17.95), 9.30 (7.75–11.18), 2.90 (1.87–4.05), and 2.36 (1.51–3.35). Thus, recurrent burden clearly separates the two high-volume chapters from the other two and ranks main rotor ahead of landing gear. It does not distinguish tail rotor from rotor drive, nor does it reproduce their COUR ordering, which also reflects parts cost and labour.

4.9. Stock-decision evidence: operational demand

Before the METRIC stock screen, operational usage effects on removal demand are quantified from the negative-binomial panel model (Section 3.10). Poisson overdispersion (Pearson $\chi^2/\mathrm{df} = 2.92$) motivates the NB specification ($\alpha = 16.6$). After aircraft, ATA, and month fixed effects, starts per flight hour (FH) are positively associated with removal intensity (IRR = 19.8; $p < 0.001$), whereas takeoffs per FH show an inverse association in the joint model (IRR = 0.28; $p < 0.001$). The pattern reflects collinearity between usage metrics rather than a causal scheduling lever. Month fixed effects are exploratory and are not used as register inputs. The register-relevant output is $\alpha = 16.6$, used for NB-calibrated METRIC sensitivity.

4.10. Incremental stock candidates

The stock screen ranks incremental repairable investments by benefit–cost ratio under Scenario S-B (minimum fill rate 85%). Table 13 is the primary output. Under S-B the optimiser invests $2.00 million, leaves total EBO at 2.526, reaches a minimum fill rate of 0.8659, and

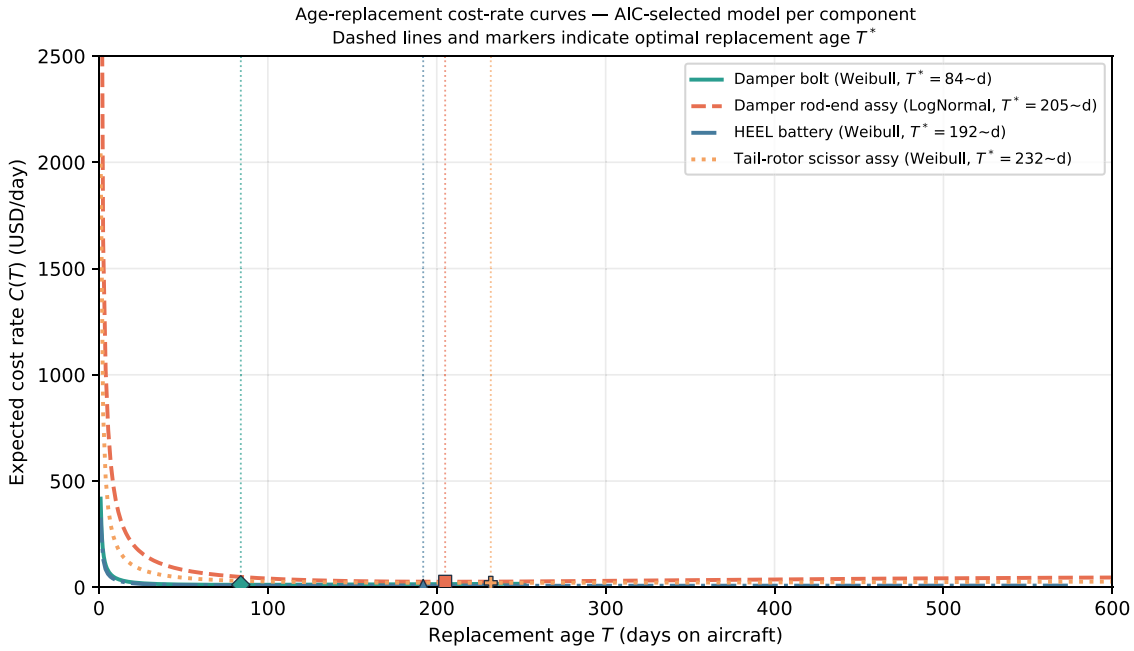


Fig. 3. Cost-rate functions $C(T)$ for the four largest-savings wear-out components, each evaluated under its AIC-selected model (specification (c); Table 11).

Table 13
Spare-parts stock recommendations under Scenario S-B (85% minimum fill rate), ranked by benefit-cost ratio. $\mathrm{BCR}_j = (\mathrm{EBO}_0 - \mathrm{EBO}_{S\text{-}B}) \times \$20,000\,\mathrm{d}^{-1} / \mathrm{investment}_j$. EBO_0: baseline expected backorders at zero stock; $\mathrm{EBO}_{S\text{-}B}$: expected backorders at recommended stock level.

Rk	Part name	Cost ($)	Stock	EBO_0	$\mathrm{EBO}_{S\text{-}B}$	Invest. ($)	BCR
1	Damper bolt	417	11	2.651	0.0003	4,584	11.57
2	HEEL battery	327	7	0.958	0.0001	2,287	8.38
3	Condenser assembly	1,144	6	0.894	0.0003	6,862	2.61
4	Tail-rotor scissor assy	2,042	8	1.821	0.0011	16,334	2.23
5	Upper scissor lever, pre-BT	3,640	11	3.418	0.0029	40,045	1.71
6	Lower scissor lever, pre-BT	3,729	6	1.246	0.0023	22,376	1.11
7	Damper rod-end assy	4,079	7	1.533	0.0016	28,553	1.07
8	Upper rod end, post-BT	5,093	7	1.853	0.0056	35,648	1.04
9	Main wheel assy, 6800 kg	17,003	8	5.813	1.3414	136,022	0.66
10	Lower scissor lever, post-BT	6,646	4	0.735	0.0050	26,582	0.55

Note: BCR values assume 1 EBO-unit of backorder reduction ≈1 aircraft-day of grounding avoided at the mid AOG day-rate. Per-part pipeline stock levels for the wider repairable set are retained in the analytical artefacts.

yields operational availability $A_o = 0.8818$; zero-stock baseline EBO is 30.696. Budget-capped alternatives (S-A $50k; S-D $200k) and the tighter fill-rate case (S-C, 95%) change investment and residual EBO but preserve the leading BCR ranks reported below. The damper bolt leads (BCR = 11.57), followed by the HEEL battery (BCR = 8.38). Eight repairables exceed BCR = 1 and are designated incremental stock candidates; the main wheel assembly remains below threshold despite its high pipeline ($D = 5.81$), indicating that lead-time reduction or process intervention would deliver greater value than stock alone.

Lead-time sensitivity (25–50 days bracketing the 35-day baseline) left the top-five BCR ranks unchanged; two mid-tier repairables fell below BCR = 1 at the shortest lead time, leaving the eight stock candidates unaffected.

The METRIC Scenario S-B sensitivity stress-tests the Poisson pipeline against NB-inflated demand ($\alpha = 16.6$ from the panel). Mean pipeline exposure D_{ij}, lead-time assumptions, AOG assumptions, and the 85% minimum fill-rate target remain unchanged. Under the panel-α NB pipeline, the same target requires $10.45 million, leaves total EBO at 6.073, reaches a minimum fill rate of 0.8503, and lowers A_o to 0.7563. The panel-α case is an upper-bound stress test, not a co-primary stock policy. Part-level weighted $\alpha = 0.32$ for the same top-20 repairables is far smaller than the panel estimate, supporting that reading; the sensitivity permits up to 500 units per repairable so that the heavy-tailed case remains feasible. The eight stock candidates are therefore calibrated against the Poisson baseline; practitioners with verified part-level α should substitute it before finalising investments.

4.11. Prospective hold-out: removal-rate validation

The hold-out tests exposure-adjusted removal priorities (Section 3.12). The development-period rate model forecasts 373.3 removals against 325 observed (+14.9%); ATA-level Spearman rank correlation is $\rho = 0.887$ across all 27 chapters, with mean absolute error 3.57 removals per chapter (root mean squared error, RMSE: 6.17). The three highest-forecast chapters (ATA 62, 32, 31) match the highest-observed hold-out chapters and the top-three COUR chapters, supporting continued economic screening on the dominant burden drivers.

4.12. Hold-out stability of COUR rankings

Hold-out parts and labour costs are available for all 386 events in the October 2025–March 2026 window (Section 3.3). Table 14 compares development and hold-out COUR for the ten leading development chapters. Split-sample rank agreement is strong (Spearman $\rho = 0.903$, $p < 0.001$; Kendall $\tau = 0.756$), and the predictive COUR forecast preserves the same rank correlation and top-three chapters (ATA 62, 32, 31). Fleet-wide hold-out COUR totals $20.2 million observed versus $23.0 million forecast (+13.9%), analogous to the removal-count layer.

Table 14
Hold-out comparison of ATA-level COUR (mid AOG scenario).

ATA	Dev. COUR ($k)	Hold-out COUR ($k)	Forecast COUR ($k)	Dev. rk	Hold-out rk	|Δrk|
62 (Main rotors)	18 398.0	3922.5	5888.7	1	1	0
32 (Landing gear)	9313.2	2703.1	2980.9	2	2	0
31 (Indicating/recording)	8620.9	2061.8	2759.3	3	3	0
63 (Rotor drive)	4911.5	699.4	1572.0	4	10	6
23 (Communications)	3472.3	1217.3	1111.4	5	7	2
34 (Navigation)	3333.1	1347.6	1066.8	6	4	2
25 (Equipment/furnishings)	2764.9	1199.9	885.0	7	8	1
72 (Engine)	2516.1	1257.9	805.4	8	6	2
24 (Electrical power)	2466.6	1274.5	789.5	9	5	4
64 (Tail rotor)	2450.4	608.0	784.3	10	11	1

Note: Development period Jan 2024–Sep 2025; hold-out Oct 2025–Mar 2026. Forecast COUR applies development-period COUR per 1,000 flight hours to hold-out exposure (8119 h). Split-sample Spearman $\rho = 0.903$; predictive Spearman $\rho = 0.903$.

Table 15
Two-dimensional continuing-airworthiness review register (mid AOG scenario).

Action type	Target	Safety disposition	Economic evidence	Review note
Engineering focus	ATA 62 (Main rotors)	Not available in the study extract; obtain an operator-approved disposition	COUR $22.3M; Δrk 0	Engineering and spares review
Engineering focus	ATA 32 (Landing gear)		COUR $12.0M; Δrk 0	Engineering and spares review
Engineering focus	ATA 31 (Indicating/recording)		COUR $10.7M; Δrk 0	Engineering and spares review
Engineering focus	ATA 72 (Engine)		COUR $3.8M; Δrk +17	Hidden-cost; bootstrap 95% CI [0, 20]
Engineering focus	ATA 23 (Communications)		COUR $4.7M; Δrk +6	High per-event parts cost
Scheduled replacement	Damper bolt		Weibull; $T^* = 84$ d; save 68.2%	Selected-model candidate
Scheduled replacement	Damper rod-end assy		Log-normal; $T^* = 205$ d; save 49.8%	Selected-model candidate
Scheduled replacement	HEEL battery		Weibull; $T^* = 192$ d; save 43.0%	Selected-model candidate
Stock candidate	Damper bolt		BCR 11.57	Incremental stock (Scenario S-B)
Stock candidate	HEEL battery		BCR 8.38	Incremental stock (Scenario S-B)
Stock candidate	Condenser assembly		BCR 2.61	Incremental stock (Scenario S-B)

Note: Safety disposition is an external, precedence-bearing input and is not inferred from ATA, cost, removal frequency, AOG, or remarks; all entries remain safety-unassessed until it is supplied. The full analytical register has 19 rows (five engineering-focus, nine finite-T^* replacement, and five highest-BCR stock candidates). This compact Mid-scenario display retains all focus rows and the three leading rows in each prescriptive group. Prescriptive rows are review candidates, not approved programme changes. Across AOG scenarios, ATA 62, 32, 31, and 72 remain in the focus set, while the second hidden-cost row changes.

Mid-tier economic ranks are less stable: ATA 63 falls from rank 4 to rank 10 while ATA 24 rises from rank 9 to rank 5 in the hold-out window. These shifts do not overturn the leading-chapter pattern but caution against treating mid-ranked COUR positions as fixed programme priorities without longer observation.

4.13. Two-dimensional continuing-airworthiness review register

Algorithm 1 closes the sequence begun in Section 3. The case study can populate the economic axis but not the safety axis of the register. The full analytical register contains 19 prioritised economic rows: five engineering-focus rows, nine replacement candidates with a finite T^*, and the five highest-BCR stock candidates. All 19 rows remain safety-unassessed pending an operator-approved disposition; no row is labelled safety-critical or non-critical from ATA membership or maintenance text. Table 15 presents a compact 11-row summary comprising all five engineering-focus rows and the three leading replacement and stock candidates. Table 15 consolidates COUR-ranked ATA chapters, selected-distribution cost-rate minima, and METRIC benefit–cost triage for engineering focus, scheduled replacement, and incremental stock. ATA 72 is included as a low-frequency, high-parts-cost screening row despite bootstrap uncertainty on the exact rank gap. The artefact is a structured reliability-review input, not an approved maintenance programme revision. A volume top-10 screen would omit ATA 72 and ATA 23, and cannot produce the replacement or stock rows unless the COUR, age-replacement, and METRIC steps are applied together. Because the safety input is unavailable for all rows, empirical concordance or conflict between the two axes remains indeterminate.

5. Discussion

The COUR workflow ranks removal economics for continuing airworthiness review by decomposing unscheduled removals into parts, labour, and scenario-bounded grounding exposure. Re-ranking on COUR can redirect engineering and continuing-airworthiness management organisation (CAMO) attention towards high per-event-cost systems that volume triage under-prioritises. Because event-level AOG duration was not observed (Section 3.3), grounding cost remains an uncertainty band. The reconstructed install base now supports right-censored, delayed-entry estimates of T^*, but these remain review candidates rather than approved intervals. Incremental stock levels likewise require demand validation. Within that scope, the register (Table 15) surfaces economically consequential systems that volume ranking hides and supports engineering focus, scheduled replacement, and incremental stock through an economic screen that cannot override the separate safety disposition. Direct numerical benchmarking is unavailable because prior studies do not report comparable linked-record COUR.

The ordering result sharpens the first contribution. Different chapter-level costs do not by themselves invalidate volume triage; re-ranking becomes decision-relevant only when a per-removal cost ratio exceeds the opposing removal-count ratio. The observed 18.5% inversion share reconciles two findings that otherwise appear contradictory: the top three chapters remain stable because their count separation dominates, while economically concentrated low-frequency chapters, especially ATA 72, move sharply. This criterion is transferable to another fleet because it uses only chapter counts and the same linked cost totals required by Eq. (1); it can therefore be evaluated before survival or stock modelling is undertaken.

5.1. Interpretation of register evidence

5.1.1. Safety boundary and non-compensatory priority

The two axes answer different questions. A high COUR identifies economic burden but does not imply high consequence severity; conversely, a low COUR cannot establish operational acceptability or lower

the priority of a required safety action. The register therefore does not aggregate safety and economics into a synthetic score. It first preserves the operator-approved safety route and then uses COUR, age-replacement, and stock evidence within the economically eligible set. In the present case study, the missing safety fields force a safety-unassessed route for every target, so the contribution is the auditable decision boundary and escalation rule rather than an estimated risk ranking.

5.1.2. COUR re-ranking and engineering-focus priorities

Removal volume is an incomplete guide for fleet reliability review. ATA chapters 62, 32, and 31 remain dominant under both volume and COUR ranking, but rank divergence elsewhere is operationally important. Bootstrap resampling (Table 9) supports the downward reordering of ATA 64 and ATA 21; the engine upward shift (ATA 72) is positive in 95% of bootstrap replicates but its 95% interval includes zero because it has only three events. Event-count reviews may therefore under-screen low-frequency, high-cost chapters even when the exact rank gap is uncertain.

The AOG leg drives the scenario variation because each unscheduled removal carries the same bounded grounding-exposure cost; it does not dominate Mid total COUR, of which parts account for 61.1%. This is a sensitivity convention, not evidence that all removals grounded the aircraft (Section 3.3). The exact full-order interval around Mid is narrow because it is bounded above by a swap between ATA 45 and ATA 33 at ranks 18–19, not by a leading-priority change. The leading-three set and ATA 72 remain in the engineering-focus screen across Low, Mid, and High, whereas the fifth focus row is scenario-dependent. CAMOs can therefore retain a four-item core review set while conditioning the secondary hidden-cost review on the local AOG valuation. Labour-rate and ALHWO-imputation sensitivity are modest by comparison, so measured grounding duration and contract-specific opportunity cost remain the first data-quality priorities.

The recorded-reason audit also bounds the interpretation of every event in the workflow. Administrative coding establishes that the final population comprises tracked, discrepancy-driven unscheduled removals recorded as unserviceable; pre-removal remarks provide specific anomaly evidence for 80.6% and no specific anomaly for the remaining 19.4%. The absence of an explicit no-fault phrase cannot be interpreted as a zero no-fault-found rate: post-removal teardown and bench-test findings are unavailable, so the shop-level no-fault-found rate cannot be estimated. Economic and recurrent-event results therefore refer to recorded removals, while the survival event indicator remains an analytical convention rather than a hardware-condition assertion.

5.1.3. Scheduled-replacement bounds

No approved interval follows directly from the cost-rate minima. The install base is now reconstructed by chaining replacement records within the same extract (Section 3.7), so T^* is no longer a lower-bound candidate derived from observed removals alone, but residual qualifications remain.

First, the reconstruction cannot recover units that were removed before the 1 January 2024 extract window opened, so a residual left-truncation bias remains for components with long characteristic lives.

Second, η under the reconstructed censoring is not uniformly longer than under the uncensored fit: for two of ten components (β_w classification unaffected) η is shorter, so "censoring elongates η" cannot be asserted as a general rule and must be read per component (Table 10).

Third, imposing Weibull on every component was unjustified. The distributional screen (Table 11) selects log-normal for five of thirteen components and exponential for one. Propagating these AIC-selected models into the T^* calculation (Table 12) changes the candidate set materially: both main wheels and the main-rotor blade no longer have a finite T^*, whereas the lower scissor lever becomes a well-founded candidate. The headline damper-bolt saving remains the largest in the set and remains Weibull-supported, but its value changes from 61.5% to 68.2% under the reconstructed specification, illustrating that the qualitative finding is more stable than any single percentage derived from it.

Subsystem recurrence complements, but cannot corroborate, the component-level survival and cost-rate evidence. The aircraft-level MCF confirms substantially greater recorded-removal burden for main rotor and landing gear than for tail rotor and rotor drive, but it does not distinguish the latter pair and should not be read as corroboration of their COUR ordering. Conversely, the 19 censoring-aware Nelson–Aalen fits preserve component-age risk sets only within component type.

5.1.4. Stock screening and hold-out stability

The BCR screen ranks incremental candidates under stated scenario bounds; it is not a full multi-item portfolio model. Lead-time sensitivity left the eight stock candidates unchanged, and the NB panel estimate justifies METRIC dispersion calibration. Hold-out COUR ranks are stable at the top ($\rho = 0.903$) but mid-tier chapters move, supporting periodic re-evaluation rather than fixed rankings. Table 15 therefore supports three actions: focus on the largest COUR chapters, review scheduled replacement for the nine components with a finite T^* under their AIC-selected model, and test incremental spares where BCR exceeds unity.

5.2. Engineering and supportability implications

The analysis supports six management actions:

1. Obtain an operator-approved safety disposition for every register target and preserve any safety-led action independently of COUR.
2. Review ATA 62 (main rotors) and ATA 32 (landing gear) early on the economic axis.
3. Review the finite-T^* age-replacement candidates in Table 12, prioritising the damper bolt and damper rod-end assembly; exclude the main wheels and main-rotor blade from age-based scheduling under their selected models.
4. Test repairables with BCR > 1 for incremental stocking.
5. Improve measured AOG-duration and ALHWO-linked labour capture, and extend the install-base reconstruction to recover removals from before the 1 January 2024 extract window where feasible.
6. Update both register axes periodically because top-tier economic priorities persisted while mid-tier ranks moved.

6. Conclusions

Linked Part 145 records can support a continuing-airworthiness review register with separate safety and economic dimensions. Under the Mid AOG scenario, fleet COUR totalled $92.21 million, and 64 of 346 unequal-count chapter pairs (18.5%) reversed their volume ordering. The leading set of ATA 62, 32, and 31 persisted across AOG scenarios, while ATA 72 remained a low-frequency, high-cost screening priority despite uncertainty in its exact rank gap.

The reconstructed installation episodes allowed the survival analysis to incorporate right censoring and delayed entry. Using each component's AIC-selected distribution in the cost-rate optimisation materially changed the candidate set: the damper bolt and damper rod-end assembly yielded the largest modelled savings (68.2% and 49.8%), whereas both main-wheel variants and the main-rotor blade had no finite optimum under their selected models. Eight repairables exceeded BCR = 1 under the baseline stock scenario. The hold-out retained the leading COUR chapters ($\rho = 0.903$), but movement among mid-tier ranks supports periodic re-evaluation rather than a fixed programme order.

These outputs are review evidence, not approved maintenance actions. Event-level grounding duration, pre-window removals, post-removal shop findings, and approved safety dispositions remain unavailable; consequently, all 19 rows in the full analytical register are safety-unassessed, and the no-fault-found rate cannot be estimated. Application to another operator requires local exposure, cost, and safety inputs together with prospective validation. Further research should extend component histories before the extract window, measure grounding duration, and test whether register-guided interventions improve availability and lifecycle cost without weakening formal safety governance. A competing-risk formulation that separates defect-driven removal from opportunity removal and life-limit retirement would also sharpen the event definition beyond what the present administrative coding supports.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools for literature organisation, drafting, and LaTeX formatting. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] Barlow RE, Proschan F. Statistical theory of reliability and life testing. New York: Holt, Rinehart and Winston; 1975.

[2] Nakagawa T. Maintenance theory of reliability. London: Springer; 2005.

[3] European Commission. Commission regulation (EU) No 1321/2014 on the continuing airworthiness of aircraft and aeronautical products, articles and equipment, and on the approval of organisations and personnel involved in these tasks. Technical report L 362, Official Journal of the European Union; 2014, As amended; Part-CAMO introduced by Commission Regulation (EU) 2020/270.

[4] Sherbrooke CC. Optimal inventory modeling of systems: Multi-echelon techniques. 2nd ed. New York: Springer; 2004.

[5] Stanton T, DeLaurentis D, Fry D. Predictive maintenance analytics and implementation for aircraft: Challenges and opportunities. Syst Eng 2023;26:216–37. http://dx.doi.org/10.1002/sys.21651.

[6] Pelt M, Stamoulis K, Apostolidis A. Data analytics case studies in the maintenance, repair and overhaul (MRO) industry. In: MATEC web of conferences. vol. 304, 2019, p. 04005. http://dx.doi.org/10.1051/matecconf/201930404005.

[7] Wang Y, Gao W, Li X, Liu Y. Joint optimization of performance-based contracting, condition-based maintenance and spare parts inventory for degrading production systems. Reliab Eng Syst Saf 2024;243:109845. http://dx.doi.org/10.1016/j.ress.2023.109845.

[8] Victoria JFC, Prasetyo YT, Chuenyindee T, Persada SF, Nadlifatin R. Building-up an aircraft reliability program: An application of failure modes and effects analysis (FMEA). In: Proceedings of the 8th international conference on industrial and business engineering. ICIBE 2022, ACM; 2022, p. 343–50. http://dx.doi.org/10.1145/3568834.3568895.

[9] Shahani AR, Babaei M. Helicopter blade reliability: Statistical data analysis and modeling. Aerosp Sci Technol 2016;55:43–8. http://dx.doi.org/10.1016/j.ast.2016.04.013.

[10] Diamoutene A, Noureddine F, Kamsu-Foguem B. Reliability analysis with proportional hazard model in aeronautics. Int J Aeronaut Space Sci 2021;22:1222–32. http://dx.doi.org/10.1007/s42405-021-00371-1.

[11] Verhoeff M, Verhagen WJC. Component maintenance planning optimization in defense aviation. Aerospace 2023;10(3):255. http://dx.doi.org/10.3390/aerospace10030255.

[12] Badía FG, Berrade MD, Lee H. An study of cost effective maintenance policies: age replacement versus replacement after N minimal repairs. Reliab Eng Syst Saf 2020;201:106949. http://dx.doi.org/10.1016/j.ress.2020.106949.

[13] Guo Y, Sun Y, Si Q, Guo X, Chen N. Probabilistic risk assessment of civil aircraft associated failures under condition-based maintenance. Reliab Eng Syst Saf 2025;253:110550. http://dx.doi.org/10.1016/j.ress.2024.110550.

[14] Nelson W. Applied life data analysis. New York: Wiley; 1982.

[15] Apostolidis A, Pelt M, Stamoulis KP. Aviation data analytics in MRO operations: Prospects and pitfalls. In: Proceedings of the annual reliability and maintainability symposium. RAMS, IEEE; 2020, p. 1–7. http://dx.doi.org/10.1109/RAMS48030.2020.9153694.

[16] Ş.ahin M, Eldemir F, Turkyilmaz A. Inventory cost minimization of spare parts in aviation industry. Transp Res Procedia 2021;59:29–37. http://dx.doi.org/10.1016/j.trpro.2021.11.094.

[17] Ghobbar AA, Mukherjee A. Optimising aircraft spare parts inventory: a cost-reduction model. IMA J Manag Math 2024. http://dx.doi.org/10.1093/imaman/dpag003, Advance article.

[18] Dendauw P, Goeman T, Claeys D, De Turck K, Fiems D, Bruneel H. Condition-based critical level policy for spare parts inventory management. Comput Ind Eng 2021;157:107369. http://dx.doi.org/10.1016/j.cie.2021.107369.

[19] Yu VF, Salsabila NY, Siswanto N, Kuo P-H. A two-stage genetic algorithm for joint coordination of spare parts inventory and planned maintenance under uncertain failures. Appl Soft Comput 2022;130:109705. http://dx.doi.org/10.1016/j.asoc.2022.109705.

[20] Zheng M, Ye H, Wang D, Pan E. Joint decisions of components replacement and spare parts ordering considering different supplied product quality. IEEE Trans Autom Sci Eng 2024;21(2):1952–63. http://dx.doi.org/10.1109/TASE.2023.3252812.

[21] Zhu S, van Jaarsveld W, Dekker R. Spare parts inventory control based on maintenance planning. Reliab Eng Syst Saf 2020;193:106600. http://dx.doi.org/10.1016/j.ress.2019.106600.

[22] Ghosh C, Maiti J, Shafiee M, Kumaraswamy KG. Reduction of life cycle costs for a contemporary helicopter through improvement of reliability and maintainability parameters. Int J Qual Reliab Manag 2018;35(2):545–67. http://dx.doi.org/10.1108/IJQRM-11-2016-0199.

[23] Satpathy N, Murari K. Life cycle cost of a six ton helicopter. J Aeronaut Soc India 2005;57:69–77.

[24] de Pater I, Reijns A, Mitici M. Alarm-based predictive maintenance scheduling for aircraft engines with imperfect remaining useful life prognostics. Reliab Eng Syst Saf 2022;221:108341. http://dx.doi.org/10.1016/j.ress.2022.108341.

[25] Duan S, Sun J, Yu Z, Liu S. Uncertain data driven predictive maintenance: a cost-oriented implementation method on aircraft system. Reliab Eng Syst Saf 2025;264:111278. http://dx.doi.org/10.1016/j.ress.2025.111278.

[26] Certa A, Enea M, Galante GM, La Fata CM. ELECTRE TRI-based approach to the failure modes classification on the basis of risk parameters: An alternative to the risk priority number. Comput Ind Eng 2017;108:100–10. http://dx.doi.org/10.1016/j.cie.2017.04.018.

[27] Gonzaga C, Ribeiro AB, Gomes VAdP, Xavier TP, Souza NBPd. Attributes and characteristics of quality in offshore air transportation service. Case Stud Transp Policy 2026. http://dx.doi.org/10.1016/j.cstp.2026.101744, Advance article.

[28] European Union Aviation Safety Agency. Easy access rules for continuing airworthiness. Technical report, European Union Aviation Safety Agency; 2025, URL: https://www.easa.europa.eu/en/document-library/easy-access-rules/online-publications/easy-access-rules-continuing-airworthiness. Revision from September 2025.

[29] SAE International. Safety assessment of general aviation airplanes and rotorcraft in commercial service. SAE aerospace recommended practice ARP5151A, SAE International; 2019, http://dx.doi.org/10.4271/ARP5151A.

[30] Hong Y, Meeker WQ, McCalley JD. Prediction of remaining life of power transformers based on left truncated and right censored lifetime data. Ann Appl Stat 2009;3(2):857–79. http://dx.doi.org/10.1214/00-AOAS231.

[31] Nelson W. Confidence limits for recurrence data—Applied to cost or number of product repairs. Technometrics 1995;37(2):147–57. http://dx.doi.org/10.1080/00401706.1995.10484299.

[32] Hilbe JM. Negative binomial regression. 2nd ed. Cambridge: Cambridge University Press; 2011.